

Tax Incidence with Endogenous Exit

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July 11, 2026

I show that the passthrough of taxes onto consumer prices is biased when agents differentially exit the observed tax base. The bias depends on the magnitude of exit and the sign of the covariance between exit and passthrough. I show how to use these inputs to bound welfare-relevant passthrough, and how that bound feeds into a correction of the marginal value of public funds (MVPF). Applied to the 2018–19 trade war, the framework bounds the passthrough of tariffs onto domestic prices between 0.86, the value implied when transshipped goods pass through none of the tariff, and the literature benchmark of full passthrough, with an exit-based MVPF ranging from 1.01 to 1.37.

JEL-Classification: H22, H26, F14

Keywords: tax incidence, sufficient statistics, tax avoidance, tariffs

^{*}Massachusetts Institute of Technology, jpmejia@mit.edu. First Draft: September 2025. I am especially grateful to Kim Ruhl and Steve Miran for extensive discussions and comments on the development of these ideas, as well as to Maxx Cook, Nathan Hendren, Andrew Johnston, and Pedro Martinez-Bruera for useful comments and suggestions. Refine.ink was used to proofread this paper for consistency and clarity.

1 Introduction

Canonically, tax incidence is governed by the relative elasticities of supply and demand, as well as market structure (Gruber 2019; Weyl and Fabinger 2013). However, two common empirical features make it difficult to assess tax incidence following a reform. First, on the supply side, agents typically differ in how much of a tax they pass through. For example, when a tax falls on firms with heterogeneous market power, they will systematically differ in how much of the tax they can pass on to consumers. Second, agents typically have the option to exit the observed tax base through, for example, avoidance, evasion, or shutdown.¹ When the two features covary, estimated passthrough reflects survivorship bias by conditioning on the agents who remain, rather than the original population. That bias has two consequences: measured passthrough is wrong, and incidence and welfare evaluations of tax instruments inherit the error. This paper provides a sufficient statistics approach to bounding the bias and applies it to the 2018 U.S.-China trade war.

Two sufficient statistics summarize the bias. The first is the share of the original tax base that exits following the reform. The second is the direction of selection: whether the agents who leave have systematically higher or lower passthrough than those who stay. While the first is generally observable, the second typically requires outside information or theoretical input. For example, when firms have labor market power, an income tax hike falls on workers; those bearing the heaviest burden are most likely to exit to the gray market. Signing that selection would require knowing something about labor market structure and the consequences of monopsony. The framework therefore leaves the direction of bias to be signed with structure and evidence rather than imposed as a maintained assumption.

These two objects turn selection from an unobserved nuisance into a bounded correction. The exit share determines the largest possible survivor-cohort gap. The sign of selection determines which side of measured passthrough contains the cohort estimand: measured passthrough is an upper bound when those who leave pass through less than those who stay, and a lower bound when they pass through more. If multiple exit margins pull in opposite directions, the framework decomposes the gap by margin and aggregates their contributions. The framework adapts the worst-case selection bounds of Manski (1989) and Manski and Pepper (2003) to a survivor-cohort setting.

The welfare stakes are immediate because the same selection problem enters both sides of the MVPF. The marginal value of public funds (MVPF) measures the welfare cost of raising a marginal dollar through a tax instrument (Mayshar 1990; Bastani 2025). Its numerator is the

1. I use *exit* for departure from the observed tax base, which in the tariff application is chiefly transshipment, or tax-base shifting, rather than the trade and industrial-organization sense of a firm or product ceasing production. True firm or product exit enters separately as the participation margin and carries different welfare implications; I keep the two named distinctly because the framework's sign turns on their opposite selection.

planner-weighted burden on each side of the market. When the planner weights the two sides asymmetrically, survivor passthrough biases that burden. A nationalist planner who values only domestic welfare inherits the bias directly. The denominator captures the fiscal externality from exit: as the tax induces more agents to leave the base, marginal revenue falls. I show that three empirical inputs bound welfare under selective exit: the exit share, the sign of selection, and the exit elasticity. I provide all three for the 2018 U.S.-China trade war.

The tariff application starts from a puzzle. A large literature using bilateral customs data finds passthrough around one for the 2018 U.S. tariffs on China, implying that U.S. importers bore almost the full burden (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020; Cavallo et al. 2021), but standard terms-of-trade logic predicts some burden on Chinese exporters. One reason these facts can coexist is that customs data condition on direct shipments. They identify passthrough among the exporters who remain in the direct China–U.S. channel, not passthrough for the cohort initially exposed to the tariff. If tariff-induced exit changes who remains in that survivor set, the near-one estimate may be a selected estimand. The question is therefore which exporters leave, and whether that bias can be signed.

Two exit margins push selection in opposite directions, and I formalize them in a partial equilibrium trade model. Exporters face Kimball (1995) demand and choose whether to ship directly or exit the observed tax base via one of two margins. The first margin is transshipment: exporters transship goods through third countries rather than ship directly from China to the United States, a response empirically documented as large by Iyoha et al. (2025) and Freund (2024). The second is a standard Melitz (2003) channel: firms facing the tariff may choose to shut down. Both margins leave the direct-shipment price regression, which conditions on products that still arrive from China. Under Kimball demand, which is standard in the trade literature (Gopinath and Itskhoki 2010; Amiti, Itskhoki, and Konings 2019) and empirically realistic (De Loecker, Eeckhout, and Unger 2020; Edmond, Midrigan, and Xu 2023), demand elasticity rises with price, so markups fall as marginal cost rises. This markup gradient also orders passthrough and exit incentives: low-cost exporters have more markup to compress, so they pass through less and gain more from leaving the direct tariff base, while high-cost exporters pass through more and are closer to shutting down. Transshipment therefore removes low-passthrough exporters from direct customs data, while participation exit removes high-passthrough exporters. The two margins move the survivor–cohort gap in opposite directions: transshipment pushes measured passthrough above the cohort value, participation exit pulls it below, and the net bias depends on which margin dominates. The empirical task is to measure the two margins on the same footing.

I measure both margins in the same bilateral customs data used by the passthrough literature. Because customs data do not identify transshipment or shutdown directly, I construct product-level classification rules in the spirit of Freund (2024). A transshipment rule identifies tariffed

products whose shipments are transshipped through third countries. A shutdown rule identifies baseline China–U.S. product flows that persistently leave the direct channel after the tariff. These rules line up closely with the firm-level evidence from Iyoha et al. (2025). By 2023, transshipment reaches 5–14% of baseline China–U.S. trade, concentrated in intermediates (7–22%) and capital goods (6–18%). Shutdown is negligible in the aggregate and comparable to transshipment only for consumption goods, where both margins are around 2%. Transshipment therefore dominates shutdown, so the model predicts that the exporters who remain are selected toward high passthrough: passthrough and survival covary positively.

I then ask whether the measured transshipment selects on passthrough in the data. Transshipment concentrates in differentiated goods, where supplier differentiation gives markups the most room to vary; this is the pattern the model predicts most sharply. A higher pre-tariff China price premium predicts more transshipment among those goods. Where measured transshipment is largest, the within-product-group unit value flips sign across the Rauch (1999) split: positive among differentiated goods, where it reads markup, and negative, though imprecise, among non-differentiated goods, where it reads cost. The flip is hard to reconcile with a cost, composition, or feasibility story, which would move both classes the same way. Products with more pre-existing hub exposure and more electronics content transship more, and bulky products less, placing the response where rerouting is feasible. An HS10 decomposition shows that, in the aggregate, the unit-value movement among surviving China shipments reflects repricing rather than a shift in the surviving variety mix. The product-level data cannot confirm the selection firm by firm, but these patterns line up with it, so I adopt the positive-covariance case as the benchmark: cohort passthrough lies below the near-one survivor estimate.

I report the correction in two steps. The worst-case exercise uses only the exit share and the sign of selection, and treats exiters as having zero passthrough. This places aggregate cohort passthrough between 0.86 and the survivor benchmark of one. Anchoring exiter passthrough at 0.55, the large-firm cost passthrough of Amiti, Itskhoki, and Konings (2019), gives a point estimate of 0.94. At the category level, the correction is largest for capital and intermediate goods, where the lower bounds are roughly 0.80. There is essentially no correction for consumption goods, consistent with firm-to-firm evidence from Flaaen, Hortaçsu, and Tintelnot (2020) and Cavallo et al. (2021). Mapping the same correction into welfare gives an aggregate exit-based marginal value of public funds (MVPF) range of roughly 1.01–1.37, with larger upper bounds for intermediate and capital goods. Applying the correction to the literature range of 1.2–1.6 (Finkelstein and Hendren 2020) moves the lower bound toward one, leaving a range broadly comparable to my exit-only estimate.

The estimates are conservative because I measure only two exit margins. Other responses can also remove liability from the observed tax base, including value misreporting (Fajgelbaum and

Khandelwal 2024), shipment splitting, and product misclassification (Fisman and Wei 2004). The welfare exercise is also deliberately narrow: it is partial equilibrium, adopts a strictly nationalist perspective, and abstracts from terms-of-trade feedbacks and further passthrough from importers to final consumers (Amiti, Itskhoki, and Konings 2019; Flaaen et al. 2025). These restrictions narrow what the tariff application estimates, not what the framework can accommodate. Wherever heterogeneous agents can leave an observed tax base, measured passthrough is a survivor object. The contribution of the paper is to make that selection explicit and show how little is needed to bound its incidence and welfare consequences.

Related Literature. This paper relates to three strands of literature, all concerned with cases where the observed empirical object differs from the policy-relevant object.

First, I contribute to the sufficient statistics literature in public finance by identifying a new selection problem for passthrough, the central sufficient statistic for incidence. That literature asks when observed responses are enough for welfare analysis. Feldstein (1999) established the elasticity of taxable income as the sufficient statistic for deadweight loss, while Chetty (2009a) showed that the same elasticity can overstate deadweight loss when avoidance involves transfers rather than real resource costs.² Weyl and Fabinger (2013) established the parallel for incidence: passthrough is sufficient for incidence under general forms of competition. I show that this logic can fail at the measurement stage. Even when passthrough is the right statistic, the passthrough measured in the data may be the wrong average. When agents exit the tax base, observed passthrough conditions on those who remain and therefore becomes a selected mean rather than the cohort object welfare requires. I do not recover the missing exiter mean with an exclusion restriction or a parametric selection model. Instead, the exit share and a sign restriction deliver worst-case bounds in the tradition of Manski (1989) and Manski and Pepper (2003): the exit share limits how large the survivor-cohort gap can be, while the sign of the covariance between passthrough and survival determines its direction. The nearest empirical antecedent is Kopczuk et al. (2016), who show that moving the collection point of a diesel tax changes both measured passthrough and revenue because evasion capacity differs across agents; the framework here turns that base-composition finding into a general survivor-to-cohort correction.

Second, I build on methodologies that infer illicit trade from official statistics. Fisman and Wei (2004) introduced the trade discrepancy approach, comparing exporter and importer reports to quantify evasion; Javorcik and Narciso (2008) extended this logic to show that evasion concentrates in differentiated products. Fisman, Moustakierski, and Wei (2008) then applied the logic to indirect trade, finding that flows via Hong Kong rise with Chinese tariffs. My approach extends the classification approach of Freund (2024), who flags products with trade patterns consistent

2. Section 2.3 distinguishes this selection correction from Chetty’s resource-cost correction to welfare.

with transshipment. I add a second set of tightening rules and a shutdown margin, and the resulting classifications line up with more granular evidence from Iyoha et al. (2025) on transshipment through Vietnam. What this activity reflects, however, is contested. Alfaro and Chor (2025) note that China’s value added in third-country exports remains substantial, raising the possibility that hub-to-U.S. surges reflect supply chain integration rather than pure transshipment. My classification rules target transshipment by requiring China-to-hub flows to scale with the hub-to-U.S. surge. If the surge reflected only independent third-country production, China-to-hub flows would not systematically track it.

Third, I reinterpret findings from the tariff-incidence literature through the survivor-cohort distinction. Studies using bilateral customs data classified by country of origin find near-complete passthrough of the 2018 tariffs (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020), but those estimates identify passthrough only among firms that continue shipping directly, excluding exporters who leave the direct China–U.S. channel through transshipment. This distinction helps explain why Cavallo et al. (2021), who use BLS microdata that tracks identical goods over time, also find near-complete passthrough at the border: their sample weights toward consumer goods, where I find essentially no transshipment. Flaaen, Hortaçsu, and Tintelnot (2020) study washing machines and show that country-specific tariffs allowed firms to relocate, muting price effects, while a global tariff closing this option caused passthrough to exceed 100 percent. That contrast illustrates the same dependence formalized here: measured passthrough depends on whether exit from the taxed channel is available.

A final comparison within this literature is Alvarez et al. (2023). They use U.S. firm-to-firm transaction data on repeated, arm’s-length intermediate-input imports and find within-match passthrough of the 2018 tariffs of 78–85 percent, below product-level estimates near one. They attribute the gap to two-sided market power: exporters facing weaker demand from dominant buyers move down their marginal-cost curves and absorb the tariff, while markup adjustments are small and offsetting. They further argue that product-level aggregates are inflated by composition toward one-off transactions. I view this as complementary. Their evidence shows that even within direct trade, estimates depend on which continuing relationships and transactions are observed. My correction addresses a different selection margin: exporters who leave direct customs data altogether through transshipment. In the paper’s notation, their estimates refine the survivor side of the comparison, while my correction moves from direct-shipment survivors toward the original exposed cohort.

2 Passthrough under Heterogeneity and Exit

Passthrough is routinely recovered by regressing prices on tax changes, and it is a key input to incidence and welfare analysis. Three empirically relevant features complicate the interpretation of the resulting coefficient. First, agents within a given tax base respond differentially to tax changes. The theoretical passthrough of a corporate tax to consumer prices, for example, varies across industries depending on a firm's market power. Second, agents typically have the option to exit the observed tax base through avoidance, evasion, or ceasing operation entirely. Third, the propensity to exit is generally correlated with the pricing response. Highly profitable multinationals with deep markups may absorb a tax hike rather than raise consumer prices, but they are also exactly the firms with the resources and incentives to route their profits through offshore tax havens, removing themselves from the observed domestic tax data. This section formalizes the resulting bias in estimated passthrough and bounds it using two sufficient statistics: the share of the market that exits and a sign restriction on the covariance between passthrough and survival.

2.1 The Selection Wedge

This subsection traces measurement bias in the passthrough coefficient from a single market through to aggregate selection. The starting point is a single submarket subject to a small tax change. Under standard assumptions on supply and demand, the tax shifts the equilibrium buyer price by the passthrough coefficient β , defined as the elasticity of the buyer price with respect to the tax wedge. Under perfect competition with a small tax, β is determined entirely by the relative magnitudes of the supply and demand elasticities, $\beta = \eta_S/(\eta_S + |\eta_D|) \in [0, 1]$ (Gruber 2019). Other market structures generalize this expression. Under oligopolistic competition, for example, β depends additionally on the curvature of demand and need not lie below one.

In practice, however, tax instruments rarely apply to a single, well-defined market. An import tariff, for example, may cover hundreds of differentiated products. Submarkets within a tax base typically differ in technology, market structure, and other primitives that load onto the curvature of supply and demand. For example, in tariff applications, submarkets may differ in foreign exporter market power, which results in systematic variation across products in how easily exporters can pass the tax onto domestic prices. As a result, estimated passthrough is typically a weighted average of these features across submarkets.

To capture this heterogeneity, consider submarkets indexed by $i \in \mathcal{I}$, distributed by a measure G . Concretely, a submarket might be a foreign exporter shipping a single variety, a firm operating in a single jurisdiction, or a product line within a firm. What unites these interpretations is that a submarket is the unit at which the passthrough coefficient may vary. Each submarket i

accordingly has its own local supply $S_i(p_i)$ and demand $D_i(p_i)$ parameterized by an underlying type ξ_i , yielding a submarket-specific passthrough coefficient

$$\beta_i \equiv \beta(\xi_i) \in [\beta_{\min}, \beta_{\max}], \quad (1)$$

where the support $[\beta_{\min}, \beta_{\max}]$ is the theoretically permissible range across submarkets given submarket primitives.

Although heterogeneity is the rule rather than the exception, empirical work rarely estimates individual β_i . Instead, researchers must aggregate over some submarkets and regress price changes on tax changes to identify an aggregate coefficient. To define this aggregate formally, let ω_i denote submarket i 's pre-tax share of total transaction value, and define $\mathbb{E}_\omega[X] \equiv \int X(i) \omega_i dG(i)$ as the share-weighted average of a submarket-level quantity. The aggregate passthrough across the original pre-tax *cohort* of submarkets is:

$$\beta_{\text{cohort}} \equiv \mathbb{E}_\omega[\beta_i]. \quad (2)$$

The empirical complication is that not every submarket remains in the observed data following the tax change. At the new post-tax state, agents in each submarket compare the payoff from continuing in the observed tax base to the best available alternative. The alternative may take the form of switching to a less-taxed channel via evasion or avoidance, ceasing operation entirely, or any other action that removes the submarket from the observed data. Let $V^d(\xi_i)$ denote the post-tax payoff from continuing and $V^a(\xi_i)$ the maximum over all such alternatives. Submarket i stays if and only if $V^d(\xi_i) \geq V^a(\xi_i)$, captured by the survival indicator:

$$\sigma_i = \mathbf{1}\{V^d(\xi_i) \geq V^a(\xi_i)\}. \quad (3)$$

The exit share is defined as $\theta \equiv 1 - \mathbb{E}_\omega[\sigma_i]$.

Because exit removes submarkets from the data, a transaction-value-weighted regression mechanically identifies the share-weighted average among *survivors*, rather than the cohort:

$$\beta_{\text{survivor}} \equiv \frac{\mathbb{E}_\omega[\beta_i \sigma_i]}{\mathbb{E}_\omega[\sigma_i]}. \quad (4)$$

The cohort estimand averages over the full pre-tax population; the survivor estimand drops submarkets with $\sigma_i = 0$. This is problematic for *ex post* policy evaluation. The relevant welfare baseline is the original cohort, but the data deliver only survivors.³

3. Both estimands hold weights fixed at the pre-tax distribution ω_i , isolating the passthrough channel. If weights are instead allowed to re-equilibrate post-tax, the share-weighted regression identifies an object that mixes

Assumption 1 (Heterogeneity and Selective Exit). (i) $\theta > 0$; (ii) $\text{Cov}_\omega(\beta_i, \sigma_i) \neq 0$.

Part (i) requires that some submarkets exit the observed tax base. Part (ii) requires that exit propensity covaries with passthrough. When either fails, $\beta_{\text{survivor}} = \beta_{\text{cohort}}$ and the framework reduces to standard incidence analysis. The empirical tariff literature illustrates both conditions. Researchers typically observe only direct trade between countries and cannot observe trade that continues indirectly via transshipment, in which exporters may transship through third countries to avoid tariffs; this generates $\theta > 0$. Moreover, the exporters with the resources and incentives to transship may systematically be those with the market power to absorb tariffs rather than pass them through, generating a nonzero covariance between σ_i and β_i , something explored in an application of this framework later in the paper.

Proposition 1 (Selection wedge). *The survivor and cohort passthrough estimands differ by the share-weighted covariance of submarket-level passthrough with survival:*

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{\mathbb{E}_\omega[\sigma_i]} = \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{1 - \theta}. \quad (5)$$

The wedge between survivor and cohort estimands factors into a magnitude and a direction. The survival share $1 - \theta$ scales the magnitude of the gap and is observable. The sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$ determines the direction and is considerably more difficult to observe. The asymmetry is fundamental: selection occurs on unobservables, so the primitive that governs β_i also governs σ_i , and neither their joint distribution nor the sign of their covariance is identified from the survivor sample. Recovering the direction therefore requires either a structural restriction or auxiliary empirical evidence, with the consequence that survivor-based estimates either overstate or understate cohort passthrough depending on which side the covariance falls. The paper develops both kinds of evidence in an application to tariffs.

Remark. Exit need not operate through a single channel. For instance, a tax may cause some firms to shut down and others to exit through evasion. When non-survival partitions into K mutually exclusive margins indexed $k = 1, \dots, K$, with $\sigma_i^{(k)} = \mathbf{1}\{i \text{ exits via margin } k\}$ and $\sum_k \sigma_i^{(k)} = 1 - \sigma_i$, the wedge of Proposition 1 decomposes additively. Letting $\theta_k \equiv \mathbb{E}_\omega[\sigma_i^{(k)}]$ and $\bar{\beta}_k \equiv \mathbb{E}_\omega[\beta_i \sigma_i^{(k)}] / \theta_k$ denote the share and average passthrough of margin- k exiters,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \sum_{k=1}^K \theta_k (\beta_{\text{survivor}} - \bar{\beta}_k). \quad (6)$$

Each term inherits the sign of $\beta_{\text{survivor}} - \bar{\beta}_k$: margins that select low-passthrough exiters contribute

passthrough with reallocation across surviving submarkets.

positively, margins that select high-passthrough exiters negatively. The aggregate sign reflects which contribution dominates.

2.2 Bounds on Cohort Passthrough

The selection wedge identifies the source of the bias but does not by itself bound the cohort estimand. Combining it with the law of total expectation does, requiring two empirical inputs (the exit share θ and the survivor passthrough β_{survivor}), the support $[\beta_{\min}, \beta_{\max}]$, and a theoretical restriction on the sign of $\text{Cov}_{\omega}(\beta_i, \sigma_i)$. Letting $\beta_{\text{exiter}} \equiv \mathbb{E}_{\omega}[\beta_i \mid \sigma_i = 0]$ denote the average passthrough among exiters, the cohort estimand decomposes as

$$\beta_{\text{cohort}} = (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{exiter}}, \quad (7)$$

so any bound on β_{exiter} delivers a bound on β_{cohort} . The remaining question is what bounds on β_{exiter} the structure of the problem permits.

Proposition 2 (Sign-restricted bound on cohort passthrough). *Suppose Assumption 1 holds and $\beta_i \in [\beta_{\min}, \beta_{\max}]$ for all i . Then*

- (i) *If $\text{Cov}_{\omega}(\beta_i, \sigma_i) > 0$, then $\beta_{\text{cohort}} \in [(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min}, \beta_{\text{survivor}}]$.*
- (ii) *If $\text{Cov}_{\omega}(\beta_i, \sigma_i) < 0$, then $\beta_{\text{cohort}} \in [\beta_{\text{survivor}}, (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\max}]$.*

The proof, in Appendix A.1.2, combines Proposition 1 with the law of total expectation.⁴ The bound operationalizes the dual structure of the wedge. The exit share θ controls the width of each interval, with tight bounds when exit is small. The sign of $\text{Cov}_{\omega}(\beta_i, \sigma_i)$ selects which interval applies, placing the cohort below the survivor estimate in case (i) and above it in case (ii). Recovering β_{cohort} does not require knowledge of which specific submarkets exited, the structural parameters generating selection, or the joint distribution of submarket-level elasticities. The sign of the covariance and the aggregate exit share suffice.

One limitation of the proposition is that it requires a sign restriction on the global covariance. This restriction may be unavailable when offsetting margins push the wedge in opposite directions. Appendix A.1.3 states a disaggregated version (Corollary 3) that imposes sign restrictions margin-by-margin instead, partitioning exit margins into those selecting low-passthrough exiters and those selecting high-passthrough exiters. The resulting interval has endpoints scaled

4. Without the sign restriction on $\text{Cov}_{\omega}(\beta_i, \sigma_i)$, the bound reduces to the worst-case interval in Manski (1989) for a bounded outcome under selective observation. The sign restriction partitions this interval at β_{survivor} and selects the relevant half. The structure parallels the monotone-treatment-selection assumption in Manski and Pepper (2003), applied to a survival-versus-cohort context rather than a treated-versus-untreated context.

by margin-class shares θ^+ and θ^- rather than the aggregate θ , and is the version invoked later in the tariff application of Section 3.

A second limitation of Proposition 2 is that the bound is worst-case. One endpoint assumes the entire exiter pool sits at the boundary of the passthrough support ($\beta_{\min}$ or $\beta_{\max}$), which is unlikely in practice. I tighten the bound by imposing modest structure on how submarkets sort across the survival margin. Under threshold selection, the marginal exiter is the submarket exactly indifferent between staying and exiting, and its passthrough is identified directly from how the survivor average varies with the exit share. The result tightens the interval by replacing the survivor-average endpoint with a structural limit.

Corollary 1 (Sharpening under threshold selection). *Suppose type ξ has a continuous distribution, σ_i is a monotonic threshold function of type, and $\beta(\cdot)$ is monotone and continuously differentiable. Let $\beta_{\text{marginal}}(\theta) \equiv \beta_{\text{survivor}}(\theta) - (1 - \theta) \frac{\partial \beta_{\text{survivor}}(\theta)}{\partial \theta}$. Then*

(i) *If $\text{Cov}_\omega(\beta_i, \sigma_i) > 0$, then $\beta_{\text{marginal}} \leq \beta_{\text{survivor}}$ and*

$$\beta_{\text{cohort}} \in \left[(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min}, (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{marginal}} \right].$$

(ii) *If $\text{Cov}_\omega(\beta_i, \sigma_i) < 0$, then $\beta_{\text{marginal}} \geq \beta_{\text{survivor}}$ and*

$$\beta_{\text{cohort}} \in \left[(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{marginal}}, (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\max} \right].$$

Each interval is contained in the corresponding interval of Proposition 2; containment is strict when $\beta(\cdot)$ is nonconstant on the survivor set.

The proof, in Appendix A.1.4, derives the tightened bounds via change of variables on the survivor integral and substitutes the result into the law of total expectation. As exit grows, the survivor average shifts because the marginal submarket leaves the sample. Differentiating the average with respect to the exit share isolates the passthrough at the boundary. Under monotone selection, the marginal exiter sets a tighter limit on the inframarginal exiter pool than the worst-case endpoint of Proposition 2. In principle, $\partial \beta_{\text{survivor}} / \partial \theta$ recovers β_{marginal} from variation in the exit share under the threshold structure. In the tariff application below, that slope is not robust enough to identify marginal-exiter passthrough, so I use the worst-case floor.⁵

Remark (The exit share alone bounds the cohort). Before any restriction on the covariance, equa-

5. Under multi-margin selection (equation (6)), each margin defines a boundary of the survivor region in type space; the empirical θ may correspond to one specific margin or to aggregate exit, and the recovered β_{marginal} should be interpreted as the marginal exiter at that margin.

tion (7) and $\beta_{\text{exiter}} \in [\beta_{\min}, \beta_{\max}]$ bound the cohort to the worst-case interval of Manski (1989),

$$\beta_{\text{cohort}} \in [(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min}, (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\max}].$$

Its width is $\theta(\beta_{\max} - \beta_{\min})$, so the exit share alone caps the survivor-cohort gap. The sign restriction of Proposition 2 selects the half on one side of β_{survivor} and Corollary 1 tightens within it, but the bound rests on the exit share alone. The interval does not narrow with more survivor data: survivor samples never identify β_{exiter} , so its width is a matter of identification, not precision.

2.3 Mapping Bias into Welfare

Sections 2.1–2.2 bound passthrough, a price object. Welfare evaluation maps that object into incidence and then into the marginal value of public funds. Incidence is the welfare burden the tax places on each side of the market. By the consumer envelope, the buyer-side first-order burden per unit pre-tax base is β_{cohort} across market structures (Weyl and Fabinger 2013); the seller-side term depends on market structure and determines the other side of the incidence calculation. I map incidence into a partial equilibrium measure of welfare using the marginal value of public funds (MVPF), the first-order welfare cost per dollar of revenue raised (Mayshar 1990; Bastani 2025).⁶ Endogenous exit enters both numerator and denominator: selection changes cohort passthrough in the incidence term, and exit erodes the tax base in the revenue term. Appendix A.3 derives the exit-based MVPF as

$$\text{MVPF}_{\text{exit}} = \frac{\lambda_B \beta_{\text{cohort}} + \lambda_S (1 - \beta_{\text{cohort}})}{(1 - \theta)(1 + \eta)}, \quad \eta \equiv -\frac{\theta}{1 - \theta} \psi \left(1 - \frac{\tau'}{\tau}\right), \quad \psi \equiv \frac{\partial \ln \theta}{\partial \ln \tau}, \quad (8)$$

where τ is the tax rate at which revenue is collected, $\tau' \leq \tau$ is the rate at which the exited base remains taxed, and λ_B and λ_S are the planner's welfare weights on buyers and sellers. For simplicity, I assume the planner applies uniform within-group weights; a richer specification would weight individuals separately, capturing heterogeneity in the social marginal utility of income.

The numerator is the planner-weighted first-order welfare cost on the cohort, $B^{\text{cohort}}[\lambda_B \beta_{\text{cohort}} + \lambda_S (1 - \beta_{\text{cohort}})]$.⁷ Welfare is assessed over everyone originally facing the tax, so exiters enter with the burden they would have borne had they remained. This keeps the correction on the same first-order incidence object as passthrough; any resource costs of exit remain outside the MVPF

6. The expression derived here abstracts from non-exit behavioral margins and forces from general equilibrium.

7. The seller term $1 - \beta_{\text{cohort}}$ is the perfect-competition benchmark; under imperfect competition the seller's first-order burden carries a markup term and the two sides need not sum to one (Weyl and Fabinger 2013), so the cancellation at $\lambda_B = \lambda_S$ holds only at that benchmark. The nationalist specification of Section 6.2 sets $\lambda_S = 0$, so neither enters the numerical results; see Appendix A.3.

expression. Replacing β_{cohort} with the survivor estimate generates bias in the numerator given

$$B^{\text{cohort}} (\lambda_B - \lambda_S)(\beta_{\text{survivor}} - \beta_{\text{cohort}}),$$

the passthrough wedge scaled by the difference in welfare weights and by the cohort base. When $\lambda_B = \lambda_S$, the buyer and seller errors cancel at the perfect-competition benchmark. When the weights differ, the buyer term rises with β_{cohort} while the seller term falls, so bounding the numerator generally requires bounding β_{cohort} from both sides. Under the nationalist specification, $\lambda_S = 0$, and the one-sided bound from Proposition 2 maps directly to the numerator.

The survivor-to-cohort adjustment of passthrough is distinct from the resource-cost correction in Chetty (2009a). In Chetty's setting, the observed elasticity can measure the relevant behavioral response, but welfare also depends on whether the cost of sheltering is a real resource cost or a transfer. The issue here is different: the first-order incidence object is still passthrough, but exit changes which population the data average over. I therefore replace β_{survivor} with bounds on β_{cohort} ; I do not price the resource costs agents incur to leave the tax base. Those costs may matter for total welfare, but they are outside the partial equilibrium exit-based MVPF in equation (8).

The denominator is the marginal revenue raised across the cohort. Because exiters may remain taxed at a lower rate τ' , revenue is $R = [(1 - \theta)\tau + \theta\tau']B^{\text{cohort}}$: a higher tax mechanically raises collections from survivors, but it also induces exit, moving units from the rate τ to the lower rate τ' . Differentiating combines these forces into the denominator of equation (8), with the fiscal externality $\eta = -(\theta/(1 - \theta))\psi(1 - \tau'/\tau)$. The elasticity ψ measures how responsive exit is to the tax, $\theta/(1 - \theta)$ scales that response by the ratio of exiters to survivors, and $1 - \tau'/\tau$ scales it by the share of the rate forgone when a unit exits. Setting $\tau' = 0$ recovers full exit and the externality $-(\theta/(1 - \theta))\psi$; in the tariff application, $\tau' = \tau^h$ for transshipment and 0 for shutdown. Thus the same elasticity has a larger revenue effect when the exit margin is large or when exit forfeits more of the rate.⁸

Within the exit channel, Propositions 1–2 and equation (8) together reduce welfare evaluation to a small set of empirical objects: the survivor passthrough β_{survivor} , the exit share θ , the exit elasticity ψ , and the sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$. The first three are estimable from standard tax data. The fourth is not, and the framework alone is silent on it. Signing the covariance requires additional structure on the economic environment that generates exit. The next section develops this structure for an application to tariffs.

8. The exit-based MVPF captures this exit margin alone; non-exit behavioral responses would add further fiscal externalities in a fuller analysis.

3 Tariff Passthrough under Endogenous Exit

The remainder of the paper applies the framework of Section 2 to an open puzzle: a large literature finds near-complete passthrough of the 2018 U.S. tariffs on China (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020; Cavallo et al. 2021), despite standard theory predicting some burden on Chinese exporters. Because these estimates rely on customs data, they condition on reported direct trade. If tariffs induce exporters to leave that channel, the estimated coefficient is survivor passthrough rather than cohort passthrough.⁹ Section 2 shows that correcting the survivor estimate requires the exit share and the sign of selection. This section develops the structure used to sign selection in the tariff setting. Exporters differ in marginal cost, face Kimball demand, and choose whether to ship directly, transship trade, or shut down. The model links passthrough heterogeneity to the two exit margins; the empirical sections then measure which margin dominates.

3.1 Environment

I work in a static partial equilibrium setting with monopolistic competition over differentiated varieties indexed by $i \in \mathcal{I}$. Each seller sets a price p_i and reaches U.S. buyers through one of two channels indexed by $r \in \{d, a\}$ (direct or transshipped), facing ad valorem tariffs $\tau^d \geq \tau^a \geq 0$. The gross-of-tax price under channel r is $\tilde{p}_i = (1 + \tau^r)p_i$.

Buyers. U.S. importers submit demand for each variety based on its gross-of-tariff price $\tilde{p}_i$ and a price aggregator $\mathbf{P}$ summarizing competing varieties within the product category,

$$q_i = D_i(\tilde{p}_i \mid \mathbf{P}).$$

$\mathbf{P}$ is exogenous at the firm level, corresponding to a standard monopolistic competition setup in which each firm is small relative to the market and takes the price distribution of rivals as given. Denoting the demand elasticity as $\varepsilon_i > 1$, I assume that demand is weakly Kimball (1995) with superelasticity

$$\kappa_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\frac{\partial \ln \varepsilon_i(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0,$$

which nests CES at $\kappa = 0$ and allows elasticity to rise with price when $\kappa < 0$. The Kimball specification is the standard non-CES generalization in the international pricing literature, motivated by evidence that markups and passthrough vary systematically with firm size and marginal

9. Firm-level studies such as part of Cavallo et al. (2021) and Flaaen, Hortaçsu, and Tintelnot (2020) avoid origin-based selection by tracking specific goods.

cost in ways constant-markup CES demand cannot generate (Gopinath and Itskhoki 2010; Amiti, Itskhoki, and Konings 2019). The full set of regularity conditions on demand, **D1–D5**, is in Appendix A.2.1.

Sellers. Each foreign exporter $i \in \mathcal{I}$ has constant marginal cost m_i , drawn independently from a continuous distribution G with support $[\underline{m}, \overline{m}] \subset (0, \infty)$ and density $g(m) > 0$. The distribution is independent of policy and of the aggregator $\mathbf{P}$. Each exporter chooses among three options: operate the direct channel at tariff τ^d , pay an additional fixed cost $F_a > 0$ to transship and operate at τ^a instead, or shut down.

Operating either channel requires paying a fixed cost $F_d \geq 0$, standard in heterogeneous-firm trade models since Melitz (2003). The transshipment fixed cost F_a is sunk and captures the expenses of establishing the alternative channel: contracting with hub-country intermediaries, repackaging or relabeling goods, and bearing the legal and reputational risk of misdeclared origin. The transshipment tariff τ^a is the expected per-unit operating cost of the indirect path, combining hub-country tariffs, enforcement risk if the transship is detected, and any quality or delay frictions. The baseline model treats F_a and τ^a as common across firms within an HS6 product; Appendix S.1.2 extends to product-specific F_a and Appendix S.1.3 to per-unit transshipment costs that depend on shipment size, and the qualitative results are preserved in both cases. Letting

$$V(m, 1 + \tau^r) \equiv \max_{p \geq m} (p - m) D((1 + \tau^r)p \mid \mathbf{P})$$

denote optimized variable profit in channel r , exporter i solves

$$\pi(m_i) = \max \{ V(m_i, 1 + \tau^d) - F_d, V(m_i, 1 + \tau^a) - F_a - F_d, 0 \}, \quad (9)$$

corresponding to direct, transship, and shutdown. Define the avoidance gain $\Delta V(m) \equiv V(m, 1 + \tau^a) - V(m, 1 + \tau^d)$.¹⁰

3.2 Pricing, Passthrough, and Selection

This subsection characterizes pricing within each channel, the resulting passthrough heterogeneity, and the partition of firms across the three options of Section 3.1.

10. I assign the transshipment decision to the exporter to preserve tractability. Transshipment could also arise from importers searching across origin countries or from trade intermediaries arbitraging tariff differentials. In Appendix S.1.1, I show that under efficient bargaining, the cutoffs and the sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$ depend only on the joint surplus across channels, independent of which side bears F_a and F_d , so the tractability choice is unimportant under standard assumptions.

Lemma 1 (Lerner condition, conditional on channel). *For each channel $r \in \{d, a\}$ with corresponding wedge τ^r , under **D1–D4**, the optimal price $p_i^*(m_i; r)$ is interior and satisfies*

$$\frac{p_i^* - m_i}{p_i^*} = \frac{1}{\varepsilon_i(\tilde{p}_i^r | \mathbf{P})}, \quad \tilde{p}_i^r = (1 + \tau^r)p_i^*.$$

Moreover: (i) p_i^* strictly increases in m_i ; (ii) p_i^* is weakly decreasing in τ^r (constant under CES); and (iii) q_i^* strictly decreases in m_i and τ^r .

Proof. See Appendix A.2.2. □

The Lerner condition delivers the standard markup inversely proportional to the demand elasticity at the delivered price. Higher marginal costs imply higher prices, lower quantities, and lower markups within each channel. The object that matters for incidence is passthrough.

Proposition 3 (Passthrough heterogeneity, conditional on channel). *Conditional on operating in the direct channel, holding $\mathbf{P}$ fixed, define passthrough with respect to the log gross tariff wedge:*

$$\beta_i \equiv \frac{d \ln \tilde{p}_i}{d \ln(1 + \tau^d)} = \frac{\varepsilon(\tilde{p}_i) - 1}{\varepsilon(\tilde{p}_i) - 1 - \kappa(\tilde{p}_i)}, \quad (10)$$

where $\tilde{p}_i = (1 + \tau^d)p_i^*(m_i)$. Under CES ($\kappa = 0$), passthrough is homogeneous across firms at one. Under Kimball ($\kappa < 0$), passthrough is increasing in marginal cost, strictly so under the strict curvature condition in Appendix A.2.1.

Proof. See Appendix A.2.3. □

The proposition writes log-wedge passthrough as the curvature factor $(\varepsilon - 1)/(\varepsilon - 1 - \kappa)$, which increases in m_i under Kimball via the Lerner condition. The economic mechanism runs through demand curvature. Low-cost exporters charge lower delivered prices, face less elastic demand, and have more markup to compress. A tariff increase therefore puts more markup at risk for them, so they absorb more of it. High-cost exporters already face more elastic demand and have less markup to compress, so they pass more of the tariff through.

With β_i heterogeneous in m_i , the partition of firms across the three options of Section 3.1 determines which firms supply the survivor sample and hence the gap between β_{survivor} and β_{cohort} . Establishing the partition requires that transshipment be expensive enough for exporters on the verge of shutting down not to finance it instead.

Assumption 2 (No re-entry). $F_a \geq \Delta V(m_p)$, where m_p is defined by $V(m_p, 1 + \tau^d) = F_d$.

Without the assumption, very-high-cost firms would transship rather than shut down, producing a non-monotone partition. The condition is vacuous when $F_d = 0$ or $\tau^a = \tau^d$. With the assumption

in place, the partition is monotone in marginal cost: low-cost firms transship, intermediate-cost firms ship directly, and high-cost firms shut down, as the next proposition formalizes.

Proposition 4 (Channel partition). *Under D1–D5, Assumption 2, and interior cutoffs, there exist unique cutoffs $\widehat{m}(\tau^d, \tau^a, F_a)$ and $m_p(\tau^d, F_d)$ with $\widehat{m} \leq m_p$ such that the cost distribution partitions almost surely as*

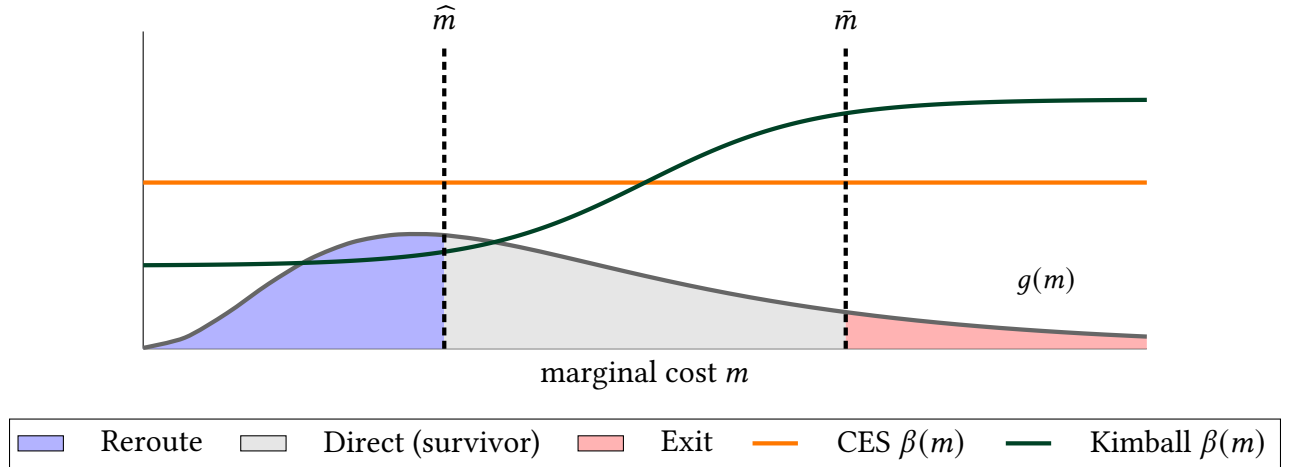
$$\begin{aligned} m < \widehat{m} &\Rightarrow \text{Transship}, \\ \widehat{m} \leq m \leq m_p &\Rightarrow \text{Direct (survivor)}, \\ m > m_p &\Rightarrow \text{Shut down}. \end{aligned}$$

At interior cutoffs, a tariff increase shrinks the survivor set from both sides:

$$\frac{\partial \widehat{m}}{\partial \tau^d} > 0, \quad \frac{\partial m_p}{\partial \tau^d} < 0.$$

The remaining cutoff sensitivities, $\partial \widehat{m} / \partial \tau^a < 0$ and $\partial m_p / \partial F_d < 0$, are derived in Appendix A.2.5. Boundary cases are interpreted with the corresponding region at zero mass.

Figure 1: Channel Partition under Heterogeneous Passthrough



Note: The distribution $g(m)$ shows the density of marginal costs. Passthrough is shown under CES (flat line) and Kimball (upward-sloping line). Dashed lines show the cutoffs at the original tariff; solid lines show the cutoffs after a tariff increase.

Figure 1 visualizes the partition. The avoidance cutoff $\widehat{m}$ is set where the avoidance gain $\Delta V(m)$ equals F_a , and the participation cutoff m_p where direct-channel variable profit $V(m, 1 + \tau^d)$

equals F_d .¹¹ The two cutoffs are driven by different forces. Low-cost exporters absorb tariffs and lose substantial surplus when tariffs rise, which they recover by paying F_a to transship at the lower wedge τ^a . High-cost exporters pass tariffs through and lose less per shipment, but their profit base is thin, so the same tariff increase pushes the marginal participant below F_d and they shut down.

The participation margin is the standard Melitz-style exit mechanism. In heterogeneous-firm trade models with fixed export costs, the least productive firms exit when costs rise. The avoidance margin, typically absent from those models, runs in the opposite direction. Formally, the two-margin partition specializes equation (6) of Section 2 to $K = 2$, with avoidance ($k = a$) selecting low-passthrough exiters and participation ($k = p$) selecting high-passthrough exiters, and the framework's prediction depends on which margin dominates.

Corollary 2 (Sign of the survivor-cohort gap). *Let $\theta^a = \Pr(m < \widehat{m})$ and $\theta^p = \Pr(m > m_p)$, and let $\bar{\beta}_a, \bar{\beta}_p$ denote the corresponding within-region average passthroughs. Then $\beta_{\text{survivor}} > \beta_{\text{cohort}}$ if and only if*

$$\theta^a(\beta_{\text{survivor}} - \bar{\beta}_a) > \theta^p(\bar{\beta}_p - \beta_{\text{survivor}}). \quad (11)$$

Proof. See Appendix A.2.6. □

The partition also pins down the structural content of the exit elasticity ψ entering the MVPF denominator of Section 2.3. With two margins operating, ψ aggregates margin-specific elasticities ψ^a and ψ^p share-weighted by their contributions θ^a/θ and θ^p/θ ; Appendix A.3 formalizes the aggregation. Both elasticities are strictly positive by Proposition 4. Holding τ^a fixed at its policy-invariant level, an increase in τ^d raises both θ^a (by widening the avoidance gap $\tau^d - \tau^a$) and θ^p (by pushing the marginal participant below F_d). When the transshipment margin dominates, $\psi \approx (\theta^a/\theta) \psi^a$, and the welfare evaluation collapses to one structural object, the elasticity of the transshipment share with respect to the avoidance gap. Greater demand curvature widens the passthrough gap between low- and high-cost exporters, amplifying the profit differential at each cutoff and raising the responsiveness of both margins.

The model therefore supplies the sign logic; the data must supply the margin sizes. Section 4 develops a methodology to measure θ^a and θ^p in customs data, and Section 5 reports the magnitudes and verifies that θ^a dominates θ^p .

11. Corollary 1 of Section 2 assumes one-sided threshold selection and so applies to each margin separately, with $\widehat{m}$ as a lower threshold and m_p as an upper threshold. The disaggregated bound of Corollary 3 combines the two.

4 Constructing Exit Measures in Customs Data

The 2018 tariffs levied by the United States on China gave Chinese exporters an incentive to exit the observed customs tax base, whether by transshipment trade through third countries or by shutting down entirely. Section 3 showed that the gap between survivor and cohort passthrough depends on this selection, but its magnitude is an empirical question. How much trade left the direct China–U.S. channel after 2018, and through which margin? This section measures both margins in customs data, which record the tax base as the taxing authority itself observes it. The two leave different footprints. Transshipped goods reappear under a third-country declared origin, so transshipment must be recovered by triangulating China–hub–U.S. flows. Shutdown leaves a one-sided decline in direct trade with no offsetting hub flow. I first describe the customs data and tariff wedge, then map the firm-level objects from the model to product-level measures, and finally construct the two exit margins.

4.1 Data

I use annual data on bilateral trade flows from CEPII’s BACI HS-12 vintage from 2012–2023 at the 6-digit product (HS6) level.¹² For each year t , country pair (i, j) , and HS6 code k , I observe the customs value of shipments $v_{i \rightarrow j, k, t}$. These flows are used to measure two forms of exit from direct China–U.S. trade. For transshipment, I look for goods that originate in China, flow to a hub country, and then arrive in the United States under a third-country origin. For shutdown, I look for persistent declines in direct China–U.S. trade that are not matched by offsetting hub flows. I restrict potential hubs to 36 countries selected for geographic proximity to China, port infrastructure, and pre-tariff trade relationships with both the U.S. and China (Table B.1). Both measurement procedures are discussed in greater detail below. I deflate all values to 2017 dollars using the BLS end-use import price index.

The trade flows identify where activity moves; the tariff wedge measures the incentive to move. A good imported directly from China pays the direct China tariff $\tau_{k,t}^d$. The same good transshipped through a hub enters the United States under the hub’s origin and pays the hub’s U.S. tariff $\tau_{k,t}^h$. The incentive to transship is the gap between the two rates, so I define the wedge as

$$\Delta_{k,t} \equiv \log \frac{1 + \tau_{k,t}^d}{1 + \tau_{k,t}^h}.$$

12. BACI provides consistent bilateral flows at annual frequency with comprehensive country coverage. Monthly UN Comtrade data have gaps, reporting lags, and reconciliation issues across partner countries that would complicate the screening methodology. The HS6 level is the finest internationally harmonized classification in bilateral trade data; finer classifications are country-specific.

The wedge depends only on the tariff paid on each route. Both $\tau_{k,t}^d$ and $\tau_{k,t}^h$ vary across products, producing heterogeneity in the wedge that I develop in Appendix S.1.4. For the direct tariff, I use Amiti, Redding, and Weinstein (2020) for monthly HS10 China-specific rates through 2017, aggregated to annual HS6 using 2017 import shares as weights. From 2018–2023, I use the Global Tariff Database from Rodríguez-Clare, Ulate, and Vasquez (2025), which provides bilateral HS6 tariffs based on the methodology in Teti (2025). For the hub tariff $\tau_{k,t}^h$, I use the U.S. tariff faced by the candidate hub countries on each HS6 product, drawn from the same Global Tariff Database and value-weighted across hubs by their pre-tariff shares of U.S. imports.¹³

I classify each HS6 product as consumption, intermediate, or capital using the UN’s Broad Economic Categories (BEC), which define the end-use categories used throughout. Table C.1 reports the resulting wedge by end-use category. In the value-weighted data used for the main estimates, the post-minus-pre increase in $\Delta_{k,t}$ is largest for intermediates (0.140), followed by capital goods (0.114) and consumption goods (0.042). The differences across end-use categories reflect the underlying direct and hub tariffs, with intermediates facing the highest direct rates.

Products that span multiple BEC categories are included once per category, with regression weights equal to the product’s BEC category share multiplied by its share of 2017 China-U.S. imports. Each product then contributes to every relevant end-use category in proportion to its composition. These classifications define the product-year cells and weights used in the empirical exercise. I next translate the firm-level exit objects from Section 3 into these product-level cells.

4.2 From Firms to Products

The model develops firm-level objects, but the customs data are organized at the product level. The data observe products as HS6×origin cells. I interpret the model’s firm i as a China-origin variety within the China cell, and $\sigma_i = 0$ as its departure from direct China–U.S. shipment through transshipment or shutdown.

As in Section 3, each firm takes the product-level price aggregator $\mathbf{P}$ as given, so passthrough depends on marginal cost alone and exit shifts which firms survive without repricing those that remain. The survivor–cohort gap therefore reflects selection, not the strategic response to thinner competition that oligopolistic passthrough would generate. Each product k consists of a continuum of firms drawing marginal costs from a continuous distribution G_k with density $g_k > 0$. Product k faces tariffs τ_k^d in the direct channel and τ_k^a in the transshipment channel, with fixed costs F_a and F_d . By Proposition 4, firms partition into three regions at the avoidance cutoff $\hat{m}_k \equiv \hat{m}(\tau_k^d, \tau_k^a, F_a)$ and the participation cutoff $m_{p,k} \equiv m_p(\tau_k^d, F_d)$. The product-level transship-

13. The Global Tariff Database provides bilateral HS6 tariffs from 2018 onward. I hold $\tau_{k,t}^h$ at its 2018 value before 2018; pre-war most-favored-nation parity between China and the candidate hubs makes the pre-period direct-minus-hub gap negligible.

ment share is $\theta_k^a \equiv G_k(\widehat{m}_k)$, the participation share is $\theta_k^p \equiv 1 - G_k(m_{p,k})$, the survivor passthrough is $\beta_{\text{survivor},k} \equiv \mathbb{E}_{G_k}[\beta(m) \mid \widehat{m}_k \leq m \leq m_{p,k}]$, and the cohort passthrough is $\beta_{\text{cohort},k} \equiv \mathbb{E}_{G_k}[\beta(m)]$.

The framework from Section 2 applies within each product, with $\theta_k = \theta_k^a + \theta_k^p$ as the within-product exit share. The sign of the within-product covariance, and hence the direction of the bound, is supplied by the model and assessed empirically in Section 5.2. Aggregating across products with 2017 import shares as weights ω_k yields share-weighted aggregates $\beta_{\text{cohort}} \equiv \mathbb{E}_{\omega}[\beta_{\text{cohort},k}]$ and $\beta_{\text{survivor}} \equiv \mathbb{E}_{\omega}[\beta_{\text{survivor},k}]$.

I need three empirical inputs at the product level: θ_k^a , θ_k^p , and $\beta_{\text{survivor},k}$. The first two come from the transshipment and shutdown measures below; the third is taken from customs-based price regressions in the literature (Amiti, Redding, and Weinstein 2019; Fajgelbaum et al. 2020; Cavallo et al. 2021). The firm-level distribution G_k and the cutoffs $(\widehat{m}_k, m_{p,k})$ are not separately identified, and within-product heterogeneity in $\beta(m)$ is unobserved. A product-level regression identifies the structural $\beta_{\text{survivor},k}$ only if the within-cell variety mix is stable; Section 5.2 examines this channel directly.

4.3 Constructing the Exit Share

The exit share $\theta = \theta^a + \theta^p$ has two components, each measured separately. Transshipment captures trade transshipped through hubs. Shutdown captures persistent below-trend direct trade with no transshipment footprint. The subsections below construct each margin in turn.

4.3.1 The Transshipped Share

Measuring transshipment is inherently difficult. Unlike domestic tax evasion, it leaves no audit trail or third-party report. I construct bounds on θ^a by screening bilateral trade data for patterns consistent with Chinese goods flowing through third countries en route to the United States. The approach extends Freund (2024), who develops conditions to identify transshipment in bilateral trade data; I tighten these conditions and calibrate thresholds to control the false positive rate. The methodology complements firm-level evidence from Vietnamese customs records (Iyoha et al. 2025) and bill-of-lading data (Do et al. 2025).

The methodology imposes five conditions that a product-hub-year triplet must satisfy simultaneously. First, the product must have faced U.S. tariff increases on China, which restricts attention to goods with a policy-induced incentive to transship. Second, China’s share of U.S. imports must have fallen below its pre-trend while a hub country’s share rose above its own, a reallocation pattern consistent with transshipment. Third, China’s exports to the rest of the world must have remained stable relative to pre-trends, ruling out the alternative that China lost global competitiveness in the product. Fourth, the hub must face a lower U.S. tariff than China on that specific

product, ensuring an economic incentive to transship through that hub. Fifth, China-to-hub trade flows must be large enough to plausibly supply the hub-to-U.S. surge. If Vietnamese firms were expanding domestic production rather than re-exporting Chinese goods, Chinese shipments to Vietnam would show no corresponding increase for the same products in the same years.

The screens are stringent by design. Each condition is evaluated against product-specific pre-trends estimated from 2012–2017, so a triplet is flagged only when trade patterns deviate sharply from historical norms. I construct two screens. The low screen flags only above-trend growth on both legs of the route; the high screen also captures scaling of pre-existing routes. Both are calibrated against a placebo sample of products that never faced tariff increases. I select thresholds such that the implied transshipment share among these never-treated products does not exceed 1% in any pre-tariff year. This calibration prioritizes specificity over sensitivity, minimizing false positives at the cost of missing some genuine transshipment.

The measure θ^a captures trade that left the direct China-U.S. channel and reappears in matched hub flows. Pure participation exit, where a Chinese firm stops exporting with no offsetting activity, produces a one-sided decline in direct trade and is not flagged by the screens, which require both a drop in China’s U.S. share and a coincident rise in a hub’s U.S. share, backed by growth in the China-to-hub leg.

The measure is a lower bound on total avoidance: the placebo calibration above holds the overall false-positive rate under one percent, and it omits margins such as value misreporting and product misclassification. Transshipment implementation details are in Appendix B.

4.3.2 The Shutdown Share

In the model, the participation margin removes a Chinese exporter from the direct channel entirely. Firm exit is not observed in HS6 trade data, so I construct a positive product-level proxy rather than a residual. The proxy flags products whose direct China–U.S. trade falls persistently below its pre-tariff trend, whose China-to-rest-of-world exports also fall below trend, and whose decline is not accounted for by detected transshipment. A residual would absorb within-survivor declines and substitution by U.S. buyers toward non-Chinese suppliers; the positive criterion excludes both.

For each HS6 code, let $r_{k,t}$ be actual real China–U.S. trade divided by its 2012–2017 log-linear pre-trend forecast, and let $r_{k,t}^{\text{ROW}}$ be the analogous ratio for China’s exports to all non-U.S. destinations. A code is flagged as shutdown if three conditions hold. First, $r_{k,t}$ falls below a calibrated threshold ϵ_{shut}^* in 2018 and 2019, and remains below that threshold in 2023. Second, $r_{k,t}^{\text{ROW}} < 1$ in 2018, 2019, and 2023, so China’s global exports of the product also fall below trend. Third, the transshipment screens flag no transshipment for the code in any post-period year. I calibrate ϵ_{shut}^* on never-treated products using the U.S.-cessation gate, holding the implied shutdown share

below one percent in every pre-tariff window; the calibrated value is $\epsilon_{\text{shut}}^* = 0.825$.

The three conditions make shutdown and transshipment disjoint by construction, so a code enters at most one of θ^a and θ^p . Because the shutdown screen carves out products flagged by the transshipment screen, the shutdown share inherits the low–high distinction; the two versions differ only in which already-transshipped codes are excluded, and they yield nearly identical shutdown shares.

The measure is a conservative proxy rather than an exact volume estimate. It assigns the full baseline value to each flagged HS6 code, so it can overstate the firm-level participation share within flagged products. It may include product-level contractions or relocation to non-hub origins when those also depress China’s rest-of-world exports, but it excludes pure within-survivor decline, U.S.-side substitution with healthy Chinese global exports, and detected transshipment. Appendix B.5 gives the implementation details.

5 Measured Exit in Customs Data

With the two exit margins constructed, I now supply the two empirical inputs the bound of Proposition 2 requires: the magnitude of exit, which scales the correction, and the sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$, which selects its direction. With an estimate of the tariff elasticity of exit, these inputs also bound exit-based welfare, which I take up in the next section.

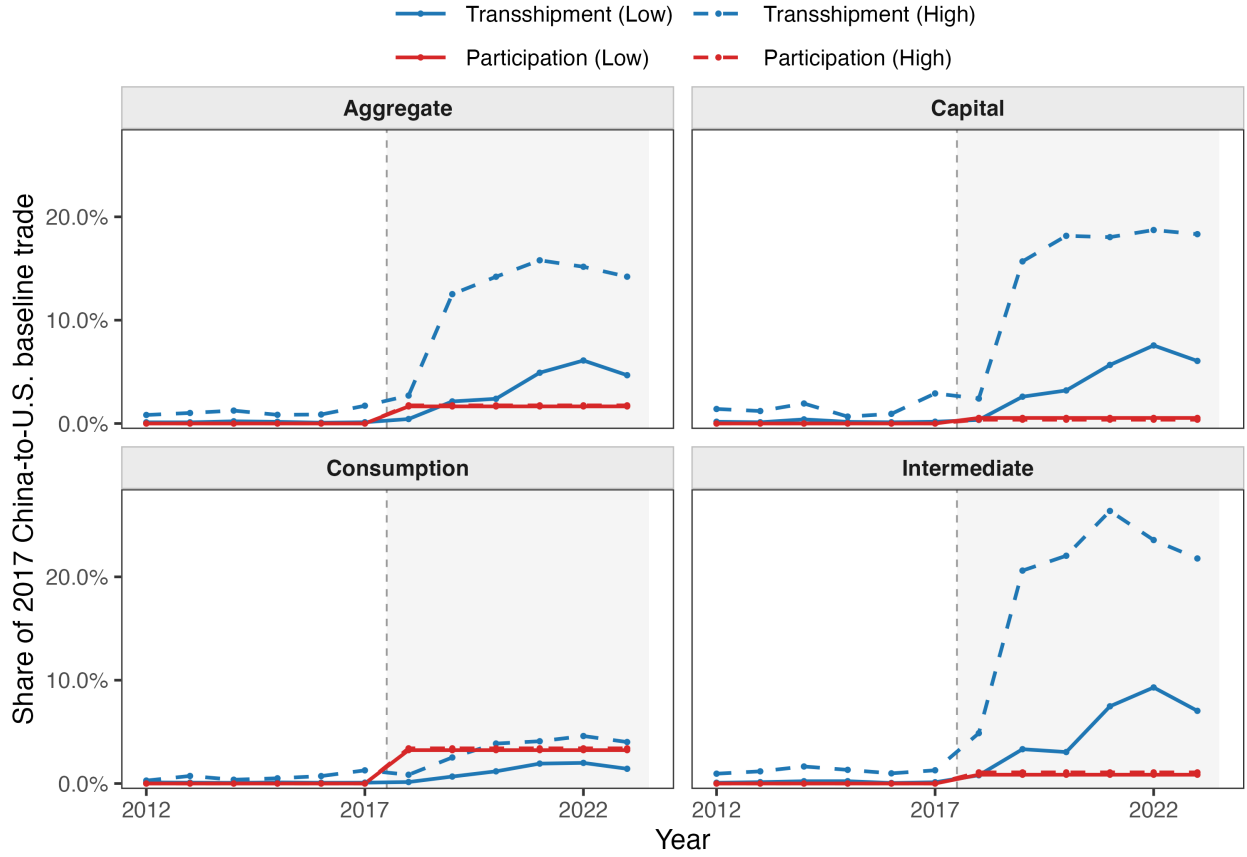
5.1 The Magnitude of Exit

Bounding passthrough starts with the size of exit after the 2018 trade war. This subsection measures it first in the time series, then exploits cross-sectional variation in tariff exposure across products to estimate the semi-elasticity of exit.

5.1.1 Exit in Levels

Figure 2 traces the magnitude of both exit margins, transshipment and shutdown, from 2012 through 2023, each as a share of 2017 baseline trade with China. Both sit near zero before 2018, when the trade war began, with no pre-existing trend. Transshipment then breaks upward with the tariffs and climbs; shutdown steps up once and stays flat. Aggregate transshipment climbs to between 5 and 14 percent of the original tax base by 2023, depending on the screen, while shutdown levels off below two percent.

Figure 2: The Two Margins of Exit by End Use



Notes: Transshipment (θ^a) and shutdown (θ^p) as a share of 2017 baseline direct China→US trade, value-weighted, among treated HS6 codes (those facing U.S. tariff increases on China). Panels show the aggregate and the three end-use categories. Solid lines are the low screen, dashed the high. See Section 4.3 for more detail.

The aggregate averages over products that faced very different tariff increases, and the two margins fall differently across them. Transshipment concentrates where the wedge is largest: under the high screen it reaches over 20 percent of baseline trade for intermediates and 18 percent for capital goods by 2023, against less than 5 percent for consumption goods, in part because consumption goods saw far milder tariff increases. Shutdown is small in all end-uses and concentrated in consumption goods, where it rises in magnitude alongside transshipment.¹⁴

The measured exit share that enters the passthrough correction rests almost entirely on transshipment rather than shutdown. Shutdown never exceeds two percent of baseline trade in the aggregate, and it is an order of magnitude smaller than transshipment in every end-use category except consumption goods. Outside evidence corroborates the transshipment magnitude: Iyoha

14. This pattern is consistent with studies that track consumption goods directly from exporter to importer, such as Flaaen, Hortaçsu, and Tintelnot (2020) and the consumption-goods analysis in Cavallo et al. (2021), and find complete passthrough: consumption goods have the smallest measured transshipment margin in these data.

et al. (2025) find that transshipment accounts for 8.8 percent of Vietnam’s growth in U.S.-destined exports over 2018–2021, while my screens imply 5.1 percent under the low screen and 14.1 percent under the high screen over the same window. At 5 to 14 percent of the tax base, exit is large enough to move measured passthrough materially. Because shutdown is small, the remaining empirical analysis focuses on transshipment, the margin that drives the aggregate exit share and the MVPF denominator.

5.1.2 The Semi-Elasticity of Transshipment

Figure 2 shows that transshipment rises when tariffs hit, but the time series alone does not show whether the rise reflects the transshipment incentive. I therefore ask whether transshipment rises more for products with larger wedge increases. I estimate

$$\theta_{k,t}^a = \alpha_k + \delta_{t \times e(k)} + \gamma \cdot \Delta_{k,t} + \varepsilon_{k,t}, \quad (12)$$

where the outcome is $\theta_{k,t}^a$ on the intensive margin or $\mathbf{1}(\theta_{k,t}^a > 0)$ on the extensive margin, α_k are HS6 fixed effects, $\delta_{t \times e(k)}$ are year $\times$ end-use fixed effects, observations are weighted by 2017 China $\rightarrow$ US import shares, and standard errors are clustered by HS6. The coefficient $\gamma \equiv \partial \theta^a / \partial \Delta$ is the semi-elasticity of transshipment with respect to the tariff wedge.¹⁵

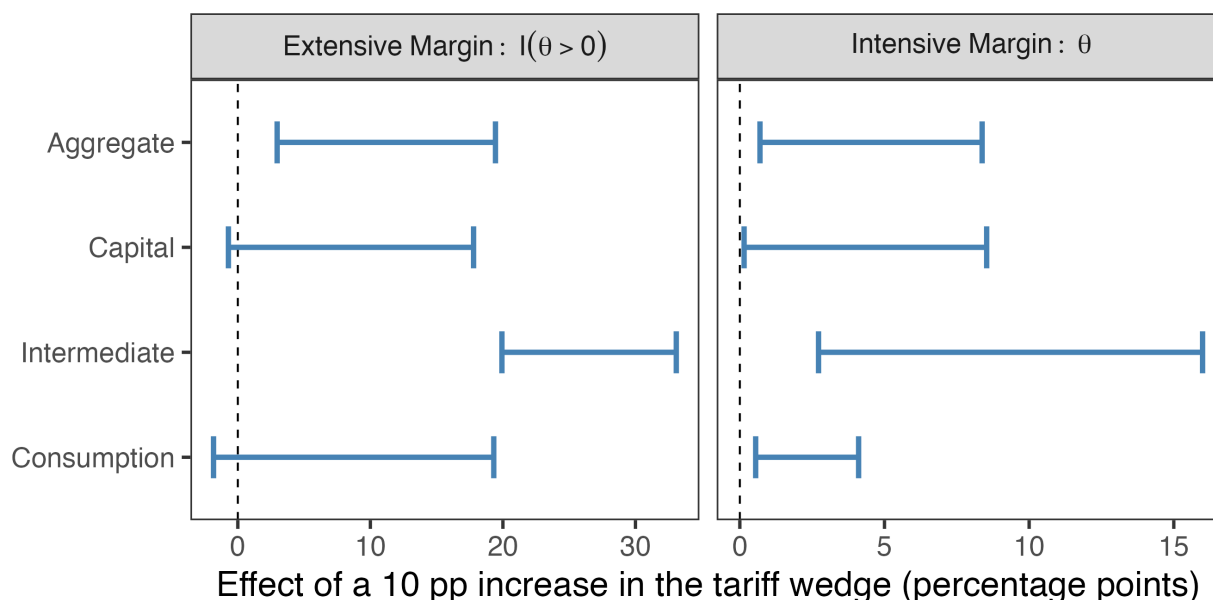
The identifying variation comes from products with larger and smaller wedge changes within an end-use-year, after absorbing permanent product differences. HS6 effects absorb time-invariant product differences. Year $\times$ end-use effects absorb shocks common to an end-use category in a given year, including the broad path of the trade war. The identifying assumption is that products receiving larger wedge increases were not already trending toward more transshipment for unrelated reasons. I report Imbens and Manski (2004) joint 95% confidence sets throughout, taking account of both the low and high variants of the transshipment measure.

I evaluate two outcomes. First, I estimate the intensive margin response of transshipment to the tariff wedge. Second, I evaluate the extensive margin $\mathbf{1}(\theta_{k,t}^a > 0)$, which asks whether tariffs cause a product to transship at all. Figure 3 plots Imbens–Manski bounds at the aggregate level and by end-use. On the intensive margin, a 10 percentage-point increase in the wedge raises the transshipment share by 0.7 to 8.4 percentage points in aggregate; on the extensive margin, it raises the probability that a product transships at all by 3 to 19 percentage points. The aggregate masks a clear ordering across end-use which follows the same ordering as the levels in Figure 2. Intermediates respond most, with intensive margin bounds of roughly 2.7 to 16 percentage points for a ten percentage point increase in tariffs, against 0.1 to 8.5 for capital and

15. The wedge enters the regression at its 2019 value for 2020–2023; the trade-war tariff schedule was essentially unchanged after 2019, so this fixes the post-2019 timing of $\Delta_{k,t}$ without affecting its cross-sectional variation.

0.5 to 4 for consumption. In other words, business inputs transship several times as readily as consumer goods. Appendix Table C.2 reports results on both the intensive and extensive margins separately for each measure of transshipment.

Figure 3: Exit Response to the Tariff Wedge



Notes: Imbens–Manski joint 95% confidence sets for γ in equation (12), aggregate and by end-use. Left panel: extensive margin (effect on the probability of any transshipment). Right panel: intensive margin (effect on the transshipment share θ^a).

Several falsification tests support the identifying assumption. The main threat is differential pre-trends: products that later received larger wedge increases may already have been moving toward more transshipment. The event study in Appendix C.1 shows that they were not; transshipment rises only after the tariffs. A block permutation test reaches the same conclusion by reassigning treated status across products within HS4 groups and comparing the pre-period treated-placebo gap to its randomization distribution. Appendix Table C.3 shows that the observed pre-period gap lies well inside that distribution.

The remaining checks rule out two mechanical alternatives. The first threat is that the estimate reflects the distribution of wedge changes and product weights rather than the actual assignment of tariff exposure across products. I test this by randomly reassigning each product’s full wedge trajectory to a donor product 5,000 times and re-estimating the main specification. The placebo coefficients center on zero and do not overlap with the true coefficients for either screen or margin (Appendix Figure C.2). The second threat is that the screens capture a global reorientation of Chinese exports after 2018 rather than transshipment induced by U.S. tariffs. I apply the same

criteria to the European Union and Canada, neither of which was directly involved in the trade war. The post-period interactions are generally indistinguishable from zero (Appendix Table C.4), so the measured response is specific to the U.S. tariff wedge.

The Supplemental Appendix checks address three remaining threats to the semi-elasticity estimates. First, the estimates are not driven by pre-tariff imbalance or product-specific trends: entropy balancing, HS4 trends, residualizing HS6 pre-trends, long differences, and a 2016–2019 window all preserve positive coefficients close to baseline (Tables S10, S8, S6, S7, and S9). Second, the result is not an artifact of inference, fixed effects, weighting, or the wedge construction: clustering by HS4 and year, adding HS2×year fixed effects, unweighted estimation, and replacing the wedge with the direct tariff alone preserve the sign and broad pattern of the estimates (Tables S1, S4, S2, and S3). Third, the result is not carried by a single hub. Leave-one-out hub exclusions move the magnitudes most for Vietnam on the extensive margin and Mexico and Japan on the intensive margin, consistent with transshipment through specific corridors, but no exclusion changes the sign of the main coefficients (Table S11).

5.2 Signing the Covariance Between Passthrough and Survival

One-sided bounds on passthrough require a sign restriction on $\text{Cov}_\omega(\beta_i, \sigma_i)$, where σ_i indicates survival in the direct tax base. The model gives the two exit margins opposite signs: transshipment removes low-passthrough firms, while participation exit removes high-passthrough firms. Section 5.1 shows participation exit is small relative to transshipment, so the positive-covariance case is the benchmark, provided the transshipment margin selects on passthrough as the model predicts. This section provides that evidence.

Under Kimball demand, low-cost exporters have more markup to compress and lower passthrough. They also gain more from leaving the direct tariff base. Transshipment therefore removes low-passthrough firms and leaves survivors selected toward high passthrough. This mechanism should be most visible where markup variation is likely to be larger.

I first test that implication by splitting products by differentiation. I re-estimate equation (12) separately for differentiated and non-differentiated goods under the Rauch (1999) classification. Differentiated goods are less interchangeable across suppliers, so exporters in those products have more scope to charge different markups. In the model, that markup dispersion links transshipment to passthrough: high-markup exporters have more room to absorb the tariff, pass through less, and gain more from leaving the direct tariff base. Table 1 shows that pattern. The wedge coefficient is larger for differentiated goods in all four outcomes, placing the transshipment response in the product class where supplier differentiation gives markups the most room to vary.

The split in Table 1 places the response where markups vary, but it does not measure them.

Unit values approximate them, but their level is ambiguous. A high unit value can reflect a high markup or a high marginal cost, and under Kimball the two move passthrough in opposite directions: a high markup compresses passthrough, a high cost raises it. I use the Rauch classification to tell the two apart. Among differentiated goods, where price differences reflect market power and quality, a high unit value proxies a high markup; among non-differentiated goods, where prices track cost, it proxies a high marginal cost. The model then predicts opposite signs across the two classes: high unit values mark the low-passthrough exporters who transship in differentiated goods, and the high-passthrough exporters who do not in non-differentiated goods.

Table 1: Exit Response by Rauch Differentiation

	Intensive		Extensive	
	θ_{low}^a	θ_{high}^a	$1(\theta_{\text{low}}^a > 0)$	$1(\theta_{\text{high}}^a > 0)$
<i>Panel A: Differentiated goods</i>				
$\Delta_{k,t}$	0.252*** (0.038)	0.806*** (0.118)	1.645*** (0.420)	1.607*** (0.420)
N	36,324	36,324	36,324	36,324
R^2	0.428	0.505	0.503	0.497
<i>Panel B: Non-differentiated goods</i>				
$\Delta_{k,t}$	0.034 (0.028)	0.158** (0.069)	0.392* (0.213)	0.530** (0.217)
N	18,588	18,588	18,588	18,588
R^2	0.357	0.522	0.618	0.626
HS6 & Year×Use FE	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Baseline wedge regression $\theta^a = \beta \Delta_{k,t} + \text{FE} + \varepsilon$ estimated separately on differentiated (Panel A) and non-differentiated (Panel B) HS6 codes under the Rauch (1999) classification. Intensive-margin outcomes are θ_{low}^a and θ_{high}^a . Extensive-margin outcomes are indicators for whether $\theta^a > 0$ under each screen. All specifications include HS6 and year×end-use fixed effects, weight by 2017 China-to-US exposure times pre-period end-use share, and cluster standard errors at the HS6 level.

Table 2 bears this out, with each measure doing a distinct job. The clearest is the China price premium, China’s pre-tariff unit value minus the world unit value for the same product, which

nets out the product's cost level and isolates China's relative price. Conditional on tariff exposure and rerouting feasibility, differentiated products with a higher premium transship more, with no such relationship among non-differentiated products; this establishes the differentiated leg, but the premium nets out cost and so cannot itself flip sign. The sharper test comes from the within-product-group China unit value, because a cost story, a composition effect, or a feasibility advantage would all move transshipment the same way in both classes, and only selection on markup flips sign across the Rauch partition. The sign flips on the high screen, where transshipment is largest: the intensive coefficient is positive among differentiated goods, where the unit value reads markup, and negative, though imprecise, among non-differentiated goods, where it reads cost. On the low screen it is near zero in both classes. The premium also predicts transshipment in both classes on the extensive margin, but that is feasibility rather than selection, since high-value products are worth the routing cost whatever their markup; I therefore read the selection from the intensive margin.

I then ask whether transshipment responds to product characteristics that should make indirect shipment easier or harder. The three proxies are concrete: pre-2017 hub exposure captures established routes through candidate hubs, bulkiness captures the cost of adding a detour, and electronics content captures products commonly traded through hub production networks. Appendix Table C.5 re-estimates the intensive-margin specification by interacting the tariff wedge with each proxy. The signs match the transshipment interpretation: products with more pre-existing hub exposure and more electronics content transship more as the wedge rises, while bulky products transship less. Appendix Table C.6 repeats the exercise on the extensive margin; hub exposure and electronics point the same way but are less precise, while bulkiness is not consistently signed. I also ask whether products with larger pre-tariff China market shares are more likely to show transshipment, after controlling for these shipment-cost proxies. Appendix Table C.10 shows that they are on the extensive margin and in flagged value, while the share itself co-moves negatively on the low screen; I treat the pattern as descriptive because baseline China-to-U.S. value affects both economic exposure to the tariff and the probability that transshipment clears the detection screen. Appendix Table C.11 asks whether transshipment responds to the cost of the indirect path: after 2018, products facing higher hub tariffs transship less, with no differential movement before the tariffs.

The checks above support interpreting $\theta_{k,t}^a$ as transshipment, but they do not directly confirm the model's predictions. Appendix Table C.9 asks whether survivor passthrough is higher in products with more transshipment. A positive gradient would be consistent with low-passthrough firms leaving the direct sample and survivors becoming more selected toward high passthrough. The table does not deliver that gradient, and instead shows a null. This failure is informative about the limit of the product-level data. The test compares products with different transshipment fea-

sibility, tariff exposure, and variety composition, while the selection claim concerns which firms leave within a product cell. I therefore do not use cross-product survivor-passthrough gradients to identify exiter passthrough or to sign selection.¹⁶

Table 2: Pre-Tariff Unit-Value Selection by Rauch Differentiation

	Intensive Margin				Extensive Margin			
	θ_{low}^a		θ_{high}^a		$1(\theta_{low}^a > 0)$		$1(\theta_{high}^a > 0)$	
	Diff.	Non-Diff.	Diff.	Non-Diff.	Diff.	Non-Diff.	Diff.	Non-Diff.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Within-HS4 standardized log pre-tariff China unit value</i>								
Pre-Tariff Measure	−0.000 (0.003)	−0.005 (0.007)	0.014* (0.007)	−0.015 (0.013)	0.066** (0.029)	−0.006 (0.031)	0.062** (0.030)	−0.007 (0.030)
Observations	2,740	1,208	2,740	1,208	2,740	1,208	2,740	1,208
<i>Panel B. China price premium (China – world unit value)</i>								
Pre-Tariff Measure	0.018 (0.013)	0.011* (0.006)	0.063** (0.026)	0.016 (0.011)	0.126*** (0.047)	0.109*** (0.038)	0.130*** (0.047)	0.104*** (0.036)
Observations	2,894	1,290	2,894	1,290	2,894	1,290	2,894	1,290
HS4 & End-Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Cross-sectional selection test. The outcome is the 2018–2023 average of the transshipment share (intensive margin, columns 1–4) or an indicator for any transshipment (extensive margin, columns 5–8), one observation per HS6×end-use cell. Each cell reports the coefficient on a pre-tariff (2012–2017) unit-value measure: within-HS4 standardized log China unit value (Panel A) or the China price premium relative to the world unit value (Panel B). Estimated separately on differentiated (‘Diff.’) and non-differentiated (‘Non-Diff.’) HS6 codes under the Rauch (1999) classification. All specifications control for post-period tariff exposure, pre-tariff China import share, China-to-hub routing intensity, bulkiness, origin concentration, and an electronics indicator; include HS4 and end-use fixed effects; weight by 2017 China→US exposure times pre-period end-use share; and cluster by HS6.

In short, the evidence indicates that the covariance between survival and passthrough is pos-

16. One concern is that the HS6 unit values used in this test move because the mix of HS10 varieties changes rather than because continuing varieties reprice. I test this by decomposing the HS6 unit-value change into within-variety repricing, composition, and a cross term in Appendix Table C.7. The result does not support a mechanical composition explanation: the aggregate movement is mostly within-variety repricing, and in capital and intermediate goods, where transshipment is largest, composition raises average unit values rather than mechanically lowering them. Appendix Table C.8 reports a related dispersion check; the result is mixed.

itive for two reasons. First, the data show that transshipment is large relative to participation exit, especially in capital and intermediate goods. Second, the selection tests show the transshipment margin drawing on higher-markup products where unit values proxy markup, and the sign reversal across the Rauch partition is hard to reconcile with a simple cost or composition story. I therefore use the positive survival-covariance case as the benchmark and apply case (i) of Proposition 2. The next section carries the bound into the numerator of the marginal value of public funds.

6 Corrected Tariff Passthrough and the MVPF

This section combines exit, selection, and the tariff elasticity of exit to first make a correction to tariff passthrough, then evaluate the exit-based marginal value of public funds.

6.1 Corrected Passthrough

With the level of exit and its direction in hand, I correct the survivor passthrough estimate for the firms that left. Because the covariance is positive, cohort passthrough lies below survivor passthrough, and case (i) of Proposition 2 gives the bound. Since participation exit is negligible, I apply the disaggregated form of the bound with the participation margin set to its framework-consistent upper bound, where participation exiters are high-cost firms passing through fully, so the correction runs through transshipment alone.

To illustrate the correction, I anchor survivor passthrough at $\beta_{\text{survivor}} = 1$, the value estimated across a wide range of customs-based studies (Amiti, Redding, and Weinstein 2019; Fajgelbaum et al. 2020; Amiti, Redding, and Weinstein 2020; Cavallo et al. 2021). I then vary the one object the data leave open, the passthrough β_{exiter} the exiting submarkets would have shown had they remained, from a floor of zero up to $\beta_{\text{exiter}} = 1$, where exiters look just like survivors and the correction vanishes. The positive covariance signed in Section 5.2 places the true value in the lower part of this range, since exiters pass through less than survivors.¹⁷

17. Ideally I would use Corollary 1 to tighten the floor, replacing the zero-passthrough exiter with the marginal one recovered from how survivor passthrough shifts with the exit share, but $\partial\beta_{\text{survivor}}/\partial\theta$ is estimated too imprecisely to pin β_{marginal} .

Table 3: Implied Cohort Passthrough Under Calibrated β_{exiter}

	β_{exiter}					
	0.00	0.25	AIK	0.75	0.90	1.00
Aggregate	0.86	0.89	0.94	0.96	0.99	1.00
Capital	0.82	0.86	0.92	0.95	0.98	1.00
Intermediate	0.78	0.84	0.90	0.95	0.98	1.00
Consumption	0.96	0.97	0.98	0.99	1.00	1.00

Notes: Implied cohort passthrough is $\beta_{\text{cohort}} = (1 - \theta^a)\beta_{\text{survivor}} + \theta^a\beta_{\text{exiter}}$, with $\beta_{\text{survivor}} = 1$ and θ^a equal to the 2023 value-weighted θ_{high}^a by end-use. The AIK column sets $\beta_{\text{exiter}} = 0.55$, the large-firm cost passthrough of Amiti, Itskhoki, and Konings (2019). The table reports the transshipment-margin correction only, with participation exiters assigned $\hat{\beta}_p = 1$.

Table 3 reports the identified interval generated by the unknown transshipping-exiter passthrough. Each column fixes β_{exiter} and applies the disaggregated version of equation (7) using the measured transshipment shares. At $\beta_{\text{exiter}} = 0$, aggregate cohort passthrough is $1 - \theta^a = 0.86$. At $\beta_{\text{exiter}} = 1$, cohort passthrough returns to one. The intermediate columns show how quickly the correction narrows as transshipping exiters become more similar to survivors. Under the low screen, the same calculation gives tighter bounds.¹⁸

The correction is larger where exit concentrates. Consumption goods barely transship and stay within a few points of one at every calibration, in line with the near-complete passthrough Flaaen, Hortaçsu, and Tintelnot (2020) and Cavallo et al. (2021) find for consumer goods. Capital and intermediate goods fall to 0.91 and 0.89 at $\beta_{\text{exiter}} = 0.5$, and to 0.82 and 0.78 at the floor. Intermediates and capital transship most, with exit shares near 0.22 and 0.18 against 0.04 for consumption, so the correction is largest where exit is largest, though it runs ten to twenty points and never overturns the finding that passthrough was high.

I also report one externally anchored point estimate inside the worst-case bounds. With Kimball demand, transshippers are in the low-cost, low-passthrough region of the distribution. Since low-cost firms sell more in equilibrium, the model maps this region to larger firms. Amiti, Itskhoki, and Konings (2019) estimate that small firms pass through cost shocks almost completely, while large-firm passthrough is about 0.55, with the rest absorbed in markups.¹⁹ The “AIK” col-

18. The same accounting can be applied to other survivor anchors. For example, treating the 0.78–0.85 within-relationship estimates of Alvarez et al. (2023) as direct-trade survivor anchors for intermediates would move the corrected cohort value downward under the positive survival-covariance case.

19. AIK estimate own-cost passthrough, the response of a firm’s price to its own marginal cost. In the model, a

umn in the table reports corrected passthrough given $\beta_{\text{exiter}} = 0.55$: aggregate passthrough is 0.94, capital and intermediate passthrough are 0.92 and 0.90, and consumption passthrough remains nearly complete. In other words, the exit-based correction is modest.

6.2 The Exit-Based MVPF

With β_{cohort} bounded and the inputs to the fiscal externality η assembled, I evaluate the exit-based marginal value of public funds of the 2018 tariffs. Under a nationalist welfare specification, which sets welfare weights on Chinese exporters equal to zero and applies a uniform weight on U.S. imports, equation (8) specializes to

$$\text{MVPF}_{\text{exit}} = \frac{\beta_{\text{cohort}}}{(1 - \theta)(1 + \eta)}. \quad (13)$$

The exit elasticity ψ entering η is recovered from the Imbens–Manski bounds on $\partial\theta/\partial\Delta$ of Section 5.1.2. I evaluate τ and θ at their 2023 values for each end-use category, weighted by 2017 import shares.

Table 4: Marginal Value of Public Funds by End-Use Category (2023)

End-use	Exit share		Fiscal externality		MVPF		
	$\bar{\theta}_{\text{low}}^a$	$\bar{\theta}_{\text{high}}^a$	η_{exit} (low)	η_{exit} (high)	MVPF (low)	MVPF (mid)	MVPF (high)
Capital	0.061	0.183	-0.00	-0.18	1.00	1.14	1.50
Consumption	0.014	0.040	-0.01	-0.05	1.01	1.04	1.09
Intermediate	0.070	0.218	-0.05	-0.38	1.06	1.34	2.06
Aggregate	0.047	0.142	-0.01	-0.15	1.01	1.12	1.37

Notes: 2023 values, using the transshipment exit margin ($\theta = \theta^a$). $\bar{\theta}^a$ is the transshipment share, 2017-share-weighted, entering the surviving base $1 - \theta^a$. η_{exit} is the fiscal externality, net of the hub tariff still collected on transshipped goods, using the Imbens–Manski slope from Figure 3. The low and high columns pair the low screen with full incidence correction ($\beta_{\text{exiter}} = 0$) and the high screen with none ($\beta_{\text{exiter}} = 1$); the midpoint takes $\beta_{\text{exiter}} = 0.5$. Table A.3 adds the shutdown margin as a robustness check.

tariff is a proportional shift in delivered marginal cost, so the same curvature factor in equation (10) governs the mapping from cost shocks to prices. The size mapping uses the monotonicity of sales in inverse cost from Lemma 1. Appendix A.2.4 provides a separate consistency check showing that $\beta_{\text{exiter}} = 0.55$ implies Kimball curvature close to standard calibrations and estimates.

Table 4 reports the MVPF by end-use at 2023 exit shares, treating transshipment as the exit margin. The low and high columns are scenario endpoints, pairing the low screen with the full passthrough correction ($\beta_{\text{exiter}} = 0$) at one end and the high screen with no correction ($\beta_{\text{exiter}} = 1$) at the other. Because each endpoint moves both inputs at once, I read the result off the midpoint at $\beta_{\text{exiter}} = 0.5$. The aggregate midpoint MVPF is 1.12, with a range from 1.01 to 1.37. Consumption goods show little transshipment and remain close to the lump-sum benchmark, with a midpoint MVPF of 1.04. Capital and intermediate goods are higher, at 1.14 and 1.34, because transshipment is larger and the fiscal externality is more negative. The fiscal externality nets the hub tariff the government still collects on transshipped goods entering through third countries, so it prices only the revenue genuinely lost to exit.

Four robustness checks bound the result. First, Table A.3 adds the shutdown margin to the exit base. Doing so raises the aggregate midpoint only to 1.14, primarily because shutdown is small relative to transshipment. Second, Appendix Table A.2 examines the negative covariance case. There, cohort passthrough lies above the survivor benchmark, causing the MVPF range to rise to 1.41–1.61 under the assumption that $\beta_{\text{max}} = 2$. Third, Appendix Table A.4 decomposes the MVPF into base erosion and the fiscal externality components, with base erosion doing most of the work. That table also reports the equal-weights variant ($\lambda_S = 1$), which raises the midpoint to 1.18 but keeps the upper envelope of the MVPF. Finally, I show that the exit elasticity would need to rise by a factor of 6.5 in the aggregate for the MVPF to reach the exit-margin Laffer ceiling.

Other estimated MVPFs are in a similar range. For example, Finkelstein and Hendren (2020) estimate an MVPF of 2.05, but note that because the change was not marginal, that estimate is too large. Their adjusted MVPF, together with Jaccard (2021), places the range of MVPFs for the trade war between 1.2 and 1.6, not far from an income tax. Applying my correction reduces passthrough below the benchmark of one, so the floor of that range falls toward one while the upper end, where passthrough is already at the benchmark and no correction applies, stays near 1.6. My own exit-based estimate sits in the same territory: a corrected aggregate of 1.12, within an envelope of 1.01–1.37, even though it only accounts for exit.²⁰

7 Conclusion

This paper generalizes the canonical textbook approach to tax incidence to environments in which passthrough is heterogeneous and agents have the option to exit. In such cases, measured passthrough is biased, and the direction of the bias depends on the sign of the covariance

20. Because the tariffs were large and discrete, two factors adjust the welfare cost relative to the marginal MVPF. Adjusting for the demand response over the size of the change lowers it, as in Finkelstein and Hendren (2020). Accounting for the resource costs of avoidance raises it to the extent those costs are real rather than transfers.

between passthrough and survival. Either feature alone leaves the standard estimator unbiased: heterogeneity without exit recovers the average passthrough across all sellers, and exit under homogeneous passthrough leaves no gap between those who remain and those who leave. The bias arises from the interaction. When agents with systematically different passthrough from the survivors exit the base, the average passthrough among survivors, the object identified in the data, differs from the average across all agents originally affected by the tax.

The size of the gap is governed by the share that exits, in the spirit of Chetty (2009b): knowing the exit share and a sign restriction on who exits is enough to bound the correction, without identifying which agents avoided or estimating the structural parameters that drove them. The direction of the bias depends on which agents leave. Section 3 illustrates the flexibility in both directions in the context of an import tariff: when transshipment dominates, the firms that leave are low-cost, low-passthrough sellers, and the standard estimate overstates measured passthrough; when participation exit dominates, the firms that leave are high-cost, high-passthrough sellers, and the bias flips. The correction is the same in both cases.

The tariff application gives a modest correction, but the mechanism is broader. The clearest analog is the personal income tax, where top earners can respond to high marginal rates by moving to lower-tax jurisdictions (Kleven et al. 2020; Rauh and Shyu 2024). The same logic applies in lower-income settings, where small firms and low-skill workers may move into the informal sector to escape the tax (Ulyssea 2020). Corporate income taxes create another version of the same problem: firms can shift paper profits abroad (Tørsløv, Wier, and Zucman 2023) or relocate real activity (Giroud and Rauh 2019). The exit channel changes across settings. The bias structure does not. In each case, the gap between the observed response and the cohort response is governed by the same identity in Proposition 1.

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A Theoretical Appendix

The theoretical appendix is organized in three parts. The first proves the framework's results from Section 2, which apply across microfoundations. The second develops the structural model from Section 3, including assumptions, comparative statics, and robustness extensions. The third derives the MVPF formula from Section 6.2.

A.1 Appendix to Section 2

A.1.1 Proof of Proposition 1

Proof. By the definitions of the cohort and survivor estimands,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \frac{\mathbb{E}_{\omega}[\beta_i \sigma_i]}{\mathbb{E}_{\omega}[\sigma_i]} - \mathbb{E}_{\omega}[\beta_i].$$

Using the covariance identity $\text{Cov}_{\omega}(\beta_i, \sigma_i) = \mathbb{E}_{\omega}[\beta_i \sigma_i] - \mathbb{E}_{\omega}[\beta_i]\mathbb{E}_{\omega}[\sigma_i]$,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \frac{\text{Cov}_{\omega}(\beta_i, \sigma_i)}{\mathbb{E}_{\omega}[\sigma_i]}.$$

□

A.1.2 Proof of Proposition 2

Proof. Define $\bar{\beta}_{\text{exiter}} \equiv \mathbb{E}_{\omega}[\beta_i \mid \sigma_i = 0]$. By the law of total expectation,

$$\beta_{\text{cohort}} = (1 - \theta) \beta_{\text{survivor}} + \theta \bar{\beta}_{\text{exiter}}, \tag{A.1}$$

where $\theta = 1 - \mathbb{E}_{\omega}[\sigma_i]$.

Case (i): $\text{Cov}_{\omega}(\beta_i, \sigma_i) \geq 0$. By the wedge identity, $\beta_{\text{survivor}} \geq \beta_{\text{cohort}}$. Substituting (A.1) gives $\beta_{\text{survivor}} \geq (1 - \theta)\beta_{\text{survivor}} + \theta\bar{\beta}_{\text{exiter}}$, which simplifies to $\bar{\beta}_{\text{exiter}} \leq \beta_{\text{survivor}}$. The assumption $\beta_i \geq \beta_{\min}$ implies $\bar{\beta}_{\text{exiter}} \geq \beta_{\min}$. Substituting these inequalities into (A.1),

$$(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min} \leq \beta_{\text{cohort}} \leq \beta_{\text{survivor}}.$$

Case (ii): $\text{Cov}_{\omega}(\beta_i, \sigma_i) \leq 0$. The wedge identity gives $\beta_{\text{survivor}} \leq \beta_{\text{cohort}}$, so by the same argument $\bar{\beta}_{\text{exiter}} \geq \beta_{\text{survivor}}$. The assumption $\beta_i \leq \beta_{\max}$ implies $\bar{\beta}_{\text{exiter}} \leq \beta_{\max}$. Substituting,

$$\beta_{\text{survivor}} \leq \beta_{\text{cohort}} \leq (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\max}.$$

□

A.1.3 Disaggregated Bound Under Margin-Specific Selection

When the global covariance $\text{Cov}_\omega(\beta_i, \sigma_i)$ has ambiguous sign—for instance, when offsetting exit margins push the wedge in opposite directions and approximately cancel—Proposition 2 delivers no informative bound. The decomposition following Proposition 1 permits a margin-specific bound when the sign of $\beta_{\text{survivor}} - \bar{\beta}_k$ is known for each k .

Partition the exit margins into $K^+ \equiv \{k : \bar{\beta}_k < \beta_{\text{survivor}}\}$, $K^- \equiv \{k : \bar{\beta}_k > \beta_{\text{survivor}}\}$, and the neutral set $K^0 \equiv \{k : \bar{\beta}_k = \beta_{\text{survivor}}\}$, with shares $\theta^+ \equiv \sum_{k \in K^+} \theta_k$ and $\theta^- \equiv \sum_{k \in K^-} \theta_k$.

Corollary 3 (Disaggregated bound). *Suppose $\theta > 0$, exit decomposes into K mutually exclusive margins as in equation (6), and the sign of $\beta_{\text{survivor}} - \bar{\beta}_k$ is known for each k . Then*

$$\beta_{\text{cohort}} \in \left[(1 - \theta^+) \beta_{\text{survivor}} + \theta^+ \beta_{\min}, (1 - \theta^-) \beta_{\text{survivor}} + \theta^- \beta_{\max} \right].$$

Proof. From the cohort decomposition $\beta_{\text{cohort}} = (1 - \theta) \beta_{\text{survivor}} + \sum_k \theta_k \bar{\beta}_k$, the constraints $\bar{\beta}_k \in [\beta_{\min}, \beta_{\text{survivor}}]$ for $k \in K^+$ and $\bar{\beta}_k \in [\beta_{\text{survivor}}, \beta_{\max}]$ for $k \in K^-$ deliver the stated interval at the extreme values. The lower endpoint sets $\bar{\beta}_k = \beta_{\min}$ on K^+ and $\bar{\beta}_k = \beta_{\text{survivor}}$ on K^- , giving $(1 - \theta) \beta_{\text{survivor}} + \theta^+ \beta_{\min} + \theta^- \beta_{\text{survivor}} = (1 - \theta^+) \beta_{\text{survivor}} + \theta^+ \beta_{\min}$. The upper endpoint is symmetric. $\square$

The interval nests the cases of Proposition 2: when $K^- = \emptyset$, $\theta^+ = \theta$ and the lower endpoint reduces to $(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min}$ as in case (i); when $K^+ = \emptyset$, the upper endpoint reduces to case (ii). When both sets are nonempty, the interval contains β_{survivor} in the interior, reflecting the wider identified set under offsetting margins. The result tightens the worst-case Manski bound uniformly: replacing θ with the smaller θ^+ on the lower endpoint and with the smaller θ^- on the upper endpoint shrinks the interval on both sides whenever both margin classes are nonempty. The informational requirement is stronger than in Proposition 2: rather than a single sign on the global covariance, the analyst must know each margin's selection direction, typically delivered by the same theoretical structure that signs the global covariance margin-by-margin, as in Corollary 2.

A.1.4 Proof of Corollary 1

Proof. Let ξ_i have continuous density $f > 0$ on its support and CDF F . Without loss of generality take $\beta(\cdot)$ monotone increasing in ξ (the decreasing case is symmetric).

Case (i): low types exit. Under threshold selection with low types exiting, $\sigma_i = \mathbf{1}\{\xi_i \geq \xi^*(\theta)\}$ where the cutoff satisfies $\theta = F(\xi^*(\theta))$. Hence $d\xi^*/d\theta = 1/f(\xi^*(\theta))$.

Define

$$G(\theta) \equiv (1 - \theta) \beta_{\text{survivor}}(\theta) = \int_{\xi^*(\theta)}^{\infty} \beta(\xi) f(\xi) d\xi.$$

Differentiating the integral form,

$$\frac{dG}{d\theta} = -\beta(\xi^*(\theta)) f(\xi^*(\theta)) \frac{d\xi^*}{d\theta} = -\beta(\xi^*(\theta)).$$

Differentiating the product form,

$$\frac{dG}{d\theta} = -\beta_{\text{survivor}}(\theta) + (1 - \theta) \frac{\partial \beta_{\text{survivor}}}{\partial \theta}.$$

Equating the two expressions,

$$\beta(\xi^*(\theta)) = \beta_{\text{survivor}}(\theta) - (1 - \theta) \frac{\partial \beta_{\text{survivor}}}{\partial \theta} \equiv \beta_{\text{marginal}}(\theta).$$

Survivors satisfy $\xi_i \geq \xi^*(\theta)$. Monotonicity of β gives $\beta(\xi_i) \geq \beta(\xi^*(\theta))$ for all survivors, so

$$\beta_{\text{survivor}}(\theta) = \mathbb{E}[\beta(\xi) \mid \xi \geq \xi^*(\theta)] \geq \beta(\xi^*(\theta)) = \beta_{\text{marginal}}(\theta),$$

with strict inequality whenever β is non-constant on the survivor set.

Exiters satisfy $\xi_i < \xi^*(\theta)$, so monotonicity gives $\beta(\xi_i) \leq \beta_{\text{marginal}}(\theta)$. Combined with $\beta_i \geq \beta_{\min}$,

$$\bar{\beta}_{\text{exiter}} \equiv \mathbb{E}_{\omega}[\beta_i \mid \sigma_i = 0] \in [\beta_{\min}, \beta_{\text{marginal}}(\theta)].$$

Substituting into the law of total expectation $\beta_{\text{cohort}} = (1 - \theta) \beta_{\text{survivor}} + \theta \bar{\beta}_{\text{exiter}}$ yields

$$\beta_{\text{cohort}} \in [(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min}, (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{marginal}}].$$

This interval is contained in the Proposition 2 case (i) interval $[(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min}, \beta_{\text{survivor}}]$, with strict containment whenever $\beta_{\text{marginal}} < \beta_{\text{survivor}}$.

Case (ii): high types exit. Under threshold selection with high types exiting, $\sigma_i = 1\{\xi_i \leq \bar{\xi}(\theta)\}$ where $\theta = 1 - F(\bar{\xi}(\theta))$. Hence $d\bar{\xi}/d\theta = -1/f(\bar{\xi}(\theta))$.

Define $G(\theta) = (1 - \theta) \beta_{\text{survivor}}(\theta) = \int_{-\infty}^{\bar{\xi}(\theta)} \beta(\xi) f(\xi) d\xi$. Differentiating two ways,

$$\frac{dG}{d\theta} = \beta(\bar{\xi}(\theta)) f(\bar{\xi}(\theta)) \frac{d\bar{\xi}}{d\theta} = -\beta(\bar{\xi}(\theta)),$$

$$\frac{dG}{d\theta} = -\beta_{\text{survivor}}(\theta) + (1 - \theta) \frac{\partial \beta_{\text{survivor}}}{\partial \theta}.$$

Equating gives the same formula,

$$\beta(\bar{\xi}(\theta)) = \beta_{\text{survivor}}(\theta) - (1 - \theta) \frac{\partial \beta_{\text{survivor}}}{\partial \theta} = \beta_{\text{marginal}}(\theta).$$

Survivors satisfy $\xi_i \leq \bar{\xi}(\theta)$, so monotonicity gives $\beta_{\text{survivor}} \leq \beta_{\text{marginal}}$, with strict inequality under non-constant β on the survivor set.

Exiters satisfy $\xi_i > \bar{\xi}(\theta)$, so $\beta(\xi_i) \geq \beta_{\text{marginal}}$. Combined with $\beta_i \leq \beta_{\text{max}}$,

$$\bar{\beta}_{\text{exiter}} \in [\beta_{\text{marginal}}(\theta), \beta_{\text{max}}].$$

Substituting into the law of total expectation,

$$\beta_{\text{cohort}} \in [(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{marginal}}, (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{max}}],$$

which is contained in the Proposition 2 case (ii) interval $[\beta_{\text{survivor}}, (1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{max}}]$, with strict containment whenever $\beta_{\text{marginal}} > \beta_{\text{survivor}}$. $\square$

Remark (The three bounds as readings of the survivor-mass curve). The three constructions are readings of the single curve $G(\theta) = (1 - \theta) \beta_{\text{survivor}}(\theta)$ at the no-exit anchor $\theta = 0$. Differentiating $G'(\theta) = -\beta(\xi^*(\theta))$ again, with $d\xi^*/d\theta = 1/f(\xi^*)$,

$$G''(\theta) = -\frac{\beta'(\xi^*(\theta))}{f(\xi^*(\theta))},$$

which is ≤ 0 in case (i), where the low exiting type sits at the bottom of an increasing β , and ≥ 0 in case (ii), where high types exit. In case (i), G is concave. The three endpoints are readings of this curve at the no-exit anchor: point identification is $\beta_{\text{cohort}} = G(0)$; the Corollary 1 endpoint, $(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\text{marginal}} = G(\theta) - \theta G'(\theta)$, is the tangent at θ ; and the Proposition 2 endpoint, $\beta_{\text{survivor}} = G(\theta)/(1 - \theta)$, is the secant through $(\theta, G(\theta))$ and the everyone-exits anchor $(1, 0)$. Concavity orders them,

$$\beta_{\text{cohort}} = G(0) \leq \underbrace{G(\theta) - \theta G'(\theta)}_{\text{Cor. 1}} \leq \underbrace{\frac{G(\theta)}{1 - \theta}}_{\text{Prop. 2}},$$

This ordering reproduces the containment of Corollary 1 within Proposition 2; case (ii) reverses the inequalities. The exiter average is the secant slope on $[0, \theta]$, $\bar{\beta}_{\text{exiter}} = (G(0) - G(\theta))/\theta$. The slack the corollary leaves unsized is

$$[G(\theta) - \theta G'(\theta)] - G(0) = \theta(\beta_{\text{marginal}} - \bar{\beta}_{\text{exiter}}) = \frac{1}{2} \theta^2 \frac{\beta'(\xi^*(0))}{f(\xi^*(0))} + o(\theta^2).$$

The two targets differ in what they ask of the data. Point identification extrapolates G to $\theta = 0$; the corollary sharpening reads its local slope at the observed θ , whose empirical stand-in is the

cross-product exit-intensity gradient. That gradient is not identified once its ties are broken at random (Appendix Table C.9), so the slope cannot be pinned and the worst-case floor is what stands.

A.2 Appendix to Section 3

A.2.1 Assumptions about Demand

I impose several regularity conditions on demand:

Assumption 3 (Demand). *The demand for each variety i depends on its own delivered price $\tilde{p}_i$ and on a price aggregator $\mathbf{P}$ summarizing rivals. Fixing $\mathbf{P}$ as parametric at the variety level, the primitives satisfy:*

D1. Downward sloping own demand and outward shifts in rivals: $\frac{\partial D_i}{\partial \tilde{p}_i} < 0$ and $\frac{\partial D_i}{\partial \mathbf{P}} > 0$.

D2. Smoothness: $D_i(\cdot \mid \mathbf{P}) \in C^2$ in own price.

D3. Elastic demand: $\varepsilon_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\frac{\tilde{p}_i}{D_i} \frac{\partial D_i}{\partial \tilde{p}_i} > 1$.

D4. Profit concavity / local stability: For either channel $c \in \{d, a\}$ with delivered price $\tilde{p}_i^c$,

$$\varepsilon(\tilde{p}_i^c \mid \mathbf{P}) - 1 - \kappa(\tilde{p}_i^c \mid \mathbf{P}) > 0.$$

D5. Kimball curvature: $\kappa_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\frac{\partial \ln \varepsilon_i(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0$ (equals 0 under CES). The restriction also requires:

$$(a) \frac{\partial \kappa(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0.$$

$$(b) \kappa'(\tilde{p}) \equiv \partial \kappa(\tilde{p}) / \partial \ln \tilde{p} \geq -\frac{\varepsilon(\tilde{p})}{\varepsilon(\tilde{p})-1} \kappa(\tilde{p})^2, \text{ with strict inequality wherever } \kappa(\tilde{p}) < 0.$$

Conditions **D1–D4** are standard and ensure well-behaved monopolistic competition with elastic demand and unique profit-maximizing prices. Condition **D5** introduces Kimball-type demand, where $\kappa \leq 0$ means the demand elasticity ε is weakly increasing in price. The technical subconditions **D5(a)–(b)** ensure that passthrough is monotone in marginal cost. The empirical literature consistently finds heterogeneous markups across firms (Gopinath and Itskhoki 2010; Amiti, Itskhoki, and Konings 2019; De Loecker, Eeckhout, and Unger 2020; Edmond, Midrigan, and Xu 2023); Kimball demand is a tractable way to generate that heterogeneity.

A.2.2 Proof of Lerner Condition

Proof. Interior and FOC. By **D2–D4**, $\pi_i(\cdot)$ is strictly concave in p_i . The unique maximizer is interior and characterized by the FOC. Let the delivered price be

$$\tilde{p}_i \equiv \tilde{p}_i(p_i, \tau^r) = \begin{cases} (1 + \tau^a)p_i & \text{(transshipment channel)} \\ (1 + \tau^d)p_i & \text{(direct channel)}. \end{cases}$$

Then $\partial \tilde{p}_i / \partial p_i \in \{1 + \tau^a, 1 + \tau^d\}$, and the FOC is

$$0 = \frac{\partial \pi_i}{\partial p_i} = D_i(\tilde{p}_i) + (p_i - m_i) D'_i(\tilde{p}_i) \frac{\partial \tilde{p}_i}{\partial p_i}.$$

With $\varepsilon_i(\tilde{p}_i) \equiv -\tilde{p}_i D'_i(\tilde{p}_i) / D_i(\tilde{p}_i)$,

$$\frac{p_i^* - m_i}{p_i^*} = \frac{\tilde{p}_i}{(\partial \tilde{p}_i / \partial p_i) p_i^*} \cdot \frac{-D_i(\tilde{p}_i)}{\tilde{p}_i D'_i(\tilde{p}_i)} = \frac{1}{\varepsilon_i(\tilde{p}_i)}.$$

This is the Lerner condition.

Monotonicity in m_i . Define

$$g(p, m; \tau^r) \equiv D(\tilde{p}) + (p - m) D'(\tilde{p}) \frac{\partial \tilde{p}}{\partial p}, \quad \tilde{p} = \tilde{p}(p, \tau^r).$$

At the optimum, $g(p^*, m; \tau^r) = 0$. Partial:

$$\frac{\partial g}{\partial m} = -D'(\tilde{p}) \frac{\partial \tilde{p}}{\partial p} > 0 \quad (\mathbf{D1}), \quad \frac{\partial g}{\partial p} = 2 D'(\tilde{p}) \frac{\partial \tilde{p}}{\partial p} + (p - m) D''(\tilde{p}) \left(\frac{\partial \tilde{p}}{\partial p} \right)^2.$$

Strict concavity of profits (**D4**) implies $\partial g / \partial p < 0$ at p^* . By the implicit function theorem, $\partial p^* / \partial m = -(\partial g / \partial m) / (\partial g / \partial p) > 0$.

Monotonicity in τ^r . For the direct channel, $\tilde{p} = (1 + \tau^r)p$. Write

$$G(p, \tau^r) \equiv \frac{p - m}{p} - \frac{1}{\varepsilon((1 + \tau^r)p)} = 0.$$

Then

$$\frac{\partial G}{\partial \tau^r} = \frac{\varepsilon'((1 + \tau^r)p)}{\varepsilon((1 + \tau^r)p)^2} p \geq 0, \quad \frac{\partial G}{\partial p} = \frac{m}{p^2} + \frac{\varepsilon'((1 + \tau^r)p)}{\varepsilon((1 + \tau^r)p)^2} (1 + \tau^r).$$

Under Kimball demand, $\varepsilon'(\cdot) \geq 0$, so $\partial G / \partial \tau^r \geq 0$. Profit concavity (**D4**, equivalently $\varepsilon - 1 - \kappa > 0$)

at the optimum) implies $\partial G/\partial p > 0$. Hence, by the IFT,

$$\frac{\partial p^*}{\partial \tau^r} = -\frac{\partial G/\partial \tau^r}{\partial G/\partial p} \leq 0,$$

with equality under CES ($\varepsilon' \equiv 0$).

Quantities. In either channel $\tilde{p} = (1 + \tau^r)p^*$ and

$$\frac{d((1 + \tau^r)p^*)}{d\tau^r} = p^* + (1 + \tau^r)\frac{dp^*}{d\tau^r} = p^* \left(1 + \frac{\kappa(\tilde{p})}{\varepsilon(\tilde{p}) - 1 - \kappa(\tilde{p})} \right) = p^* \frac{\varepsilon(\tilde{p}) - 1}{\varepsilon(\tilde{p}) - 1 - \kappa(\tilde{p})} > 0,$$

since $\varepsilon(\tilde{p}) > 1$ and the denominator is positive by D4. With $D'(\cdot) < 0$, $q_i^* = D(\tilde{p})$ strictly falls in τ^r . Similarly, q_i^* strictly falls in m because p^* rises in m and demand slopes down. This proves (i)–(iii). $\square$

Lemma 1 supplies the sorting mechanism. The Lerner condition makes passthrough depend on how the wedge shifts delivered prices and how demand elasticity varies with those prices. The comparative statics then order firms by marginal cost: higher-cost firms charge higher pre-wedge prices, sell less, and become more exposed to exit when wedges differ across channels.

A.2.3 Proof of Proposition 3

Proof. Write the first-order condition for a firm in the direct channel as $G(p, \tau^d) \equiv (p - m)/p - 1/\varepsilon((1 + \tau^d)p) = 0$. Totally differentiate and use

$$\frac{\partial G}{\partial \tau^d} = \frac{\varepsilon'((1 + \tau^d)p)}{\varepsilon((1 + \tau^d)p)^2} p, \quad \frac{\partial G}{\partial p} = \frac{m}{p^2} + \frac{\varepsilon'((1 + \tau^d)p)}{\varepsilon((1 + \tau^d)p)^2} (1 + \tau^d).$$

Using $\kappa(u) = -u \varepsilon'(u)/\varepsilon(u)$ and the Lerner condition $m = p^*(\varepsilon - 1)/\varepsilon$,

$$\frac{dp^*}{d\tau^d} = -\frac{\partial G/\partial \tau^d}{\partial G/\partial p} = \frac{\kappa(\tilde{p}^d)}{1 + \tau^d} \cdot \frac{p^*}{\varepsilon(\tilde{p}^d) - 1 - \kappa(\tilde{p}^d)}.$$

Finally, the log-wedge passthrough is

$$\beta_i = \frac{d \ln((1 + \tau^d)p^*)}{d \ln(1 + \tau^d)} = 1 + \frac{1 + \tau^d}{p^*} \frac{dp^*}{d\tau^d} = \frac{\varepsilon(\tilde{p}^d) - 1}{\varepsilon(\tilde{p}^d) - 1 - \kappa(\tilde{p}^d)}.$$

Under Assumption D4, strict profit concavity at the optimum is equivalent to $\varepsilon - 1 - \kappa > 0$, which ensures the denominator is positive. Differentiating $\beta = (\varepsilon - 1)/(\varepsilon - 1 - \kappa)$ in $\ln \tilde{p}$ and using

$$\partial \varepsilon / \partial \ln \tilde{p} = -\kappa \varepsilon,$$

$$\frac{\partial \beta}{\partial \ln \tilde{p}} = \frac{\kappa^2 \varepsilon + (\varepsilon - 1) \kappa'}{(\varepsilon - 1 - \kappa)^2}.$$

Under **D5(b)**, $(\varepsilon - 1)\kappa' \geq -\varepsilon\kappa^2$, so the numerator is ≥ 0 , with strict inequality wherever $\kappa < 0$. Combined with Lemma 1, which gives $\partial \tilde{p} / \partial m > 0$, this yields $\partial \beta / \partial m \geq 0$, strict under Kimball ($\kappa < 0$) and $= 0$ under CES ($\kappa \equiv 0$). $\square$

A.2.4 Curvature Consistency of the Exiter Calibration

Equation (10) writes log-wedge passthrough as the curvature factor $\phi \equiv (\varepsilon - 1)/(\varepsilon - 1 - \kappa) = 1/(1 + |\kappa|/(\varepsilon - 1))$ directly. The exiter calibration $\beta_{\text{exiter}} = 0.55$ from Amiti, Itskhoki, and Konings (2019) fixes the curvature factor at $\phi_{\text{exiter}} \approx 0.55$, equivalently a markup elasticity $|\kappa|/(\varepsilon - 1) \approx 0.82$. This object is the superelasticity primitive of Gopinath and Itskhoki (2010), who study the same Kimball (1995) demand. Their markup elasticity $\varepsilon/(\sigma - 1)$ is exactly $|\kappa|/(\varepsilon - 1)$ in the notation here, and their flexible-price passthrough $\Psi = 1/(1 + \varepsilon/(\sigma - 1))$ is exactly ϕ (their Proposition 1). At their calibration ($\sigma = 5$, $\varepsilon = 4$; their Section 4.2), the Kimball benchmark passthrough is $\phi = 0.50$. The exiter value 0.55 corresponds to a slightly smaller superelasticity, $\varepsilon \approx 3.3$, so the calibration sits just above the standard benchmark rather than at an extreme of the curvature.

Table A.1: The Exiter Calibration Against Measured Demand Curvature

Source	Demand system	ϕ	Relation to $\beta_{\text{exiter}} = 0.55$
Gopinath–Itskhoki (2010)	Kimball	0.50	Structural identity; 0.55 at superelasticity $\varepsilon \approx 3.3$
Klenow–Willis (2016)	Kimball	0.67	Micro-data superelasticity ≈ 2 brackets 0.55 from above
Amiti–Itskhoki–Konings (2019)	Oligopoly	0.55	Calibration source (largest-firm passthrough)
Berman–Martin–Mayer (2012)	CES + dist. costs	—	Corroborates the size gradient

Notes: ϕ is the log-wedge passthrough factor in equation (10). The first two rows map Kimball superelasticity benchmarks into ϕ at $\sigma = 5$. The last two rows report external passthrough benchmarks used to assess $\beta_{\text{exiter}} = 0.55$. The cited studies measure cost or exchange-rate passthrough; the mapping treats a tariff as a proportional shift in gross-price marginal cost.

Klenow and Willis (2016) provide a nearby data-side benchmark. They summarize micro-data estimates of the superelasticity as closer to two than to ten; at $\sigma = 5$, a superelasticity of two implies curvature passthrough near 0.67. The calibration therefore lies between the

Gopinath–Itskhoki benchmark of 0.50 and the Klenow–Willis data-side benchmark of 0.67. The firm-level gradient is also consistent with Amiti, Itskhoki, and Konings (2019), whose largest-firm passthrough is the 0.55 I adopt, and with Berman, Martin, and Mayer (2012), who find that high-performance exporters absorb more of a shock in markups. Table A.1 collects these comparisons.

A.2.5 Proof of Proposition 4

Proof. By **D2–D4**, the maximizer in $V(m, 1 + \tau^r)$ is interior and unique for each $r \in \{d, a\}$; envelope arguments apply.

Monotonicity of ΔV . The primitive $(p - m)D((1 + \tau^r)p)$ has cross-partial in $(m, 1 + \tau^r)$ equal to $-p D'((1 + \tau^r)p) > 0$ by **D1**, so it has strictly increasing differences in $(m, 1 + \tau^r)$. By Topkis, V inherits strictly increasing differences, so $V_m(m, 1 + \tau^r)$ is strictly increasing in $1 + \tau^r$. Since $\tau^a < \tau^d$,

$$\Delta V_m(m) \equiv V_m(m, 1 + \tau^a) - V_m(m, 1 + \tau^d) < 0,$$

so ΔV is strictly decreasing in m .

Existence and uniqueness of cutoffs. The avoidance cutoff solves $\Delta V(\widehat{m}) = F_a$. Since ΔV is continuous and strictly decreasing, a unique $\widehat{m}$ satisfies the indifference condition whenever $F_a \in [\Delta V(m_p), \Delta V(\underline{m})]$ (the interior case). The participation cutoff solves $V(m_p, 1 + \tau^d) = F_d$. By envelope, $V_m(m, 1 + \tau^d) = -q^d(m) < 0$, so $V(\cdot, 1 + \tau^d)$ is continuous and strictly decreasing, giving a unique m_p for any $F_d \in (0, V(\underline{m}, 1 + \tau^d))$.

Ordering. By Assumption 2, $F_a \geq \Delta V(m_p)$. Combined with $\Delta V(\widehat{m}) = F_a$ and strict monotonicity of ΔV , $\widehat{m} \leq m_p$.

Partition. Let $\pi_d(m) = V(m, 1 + \tau^d) - F_d$, $\pi_a(m) = V(m, 1 + \tau^a) - F_a - F_d$, and $\pi_e = 0$.

For $m < \widehat{m}$: from $\widehat{m} \leq m_p$ and strict monotonicity of $V(\cdot, 1 + \tau^d)$, $V(m, 1 + \tau^d) > F_d$, so $\pi_d(m) > 0$; and $\Delta V(m) > \Delta V(\widehat{m}) = F_a$ gives $\pi_a(m) - \pi_d(m) = \Delta V(m) - F_a > 0$. Hence $\pi_a > \pi_d > 0 = \pi_e$, and transshipment is optimal.

For $\widehat{m} \leq m \leq m_p$: $\Delta V(m) \leq F_a$ gives $\pi_d \geq \pi_a$, and $V(m, 1 + \tau^d) \geq F_d$ gives $\pi_d \geq 0 = \pi_e$. Direct is optimal.

For $m > m_p$: $V(m, 1 + \tau^d) < F_d$ gives $\pi_d < 0$. By Assumption 2 and strict monotonicity of ΔV , $\Delta V(m) < \Delta V(m_p) \leq F_a$, so $V(m, 1 + \tau^a) < V(m, 1 + \tau^d) + F_a < F_d + F_a$, hence $\pi_a(m) < 0$. Exit is optimal.

Boundary firms are indifferent on a measure-zero set under continuous G , so the partition holds almost surely.

Comparative statics. By the implicit function theorem applied to $\Delta V(\widehat{m}) = F_a$ and $V(m_p, 1 +$

$\tau^d) = F_d$:

$$\frac{\partial \widehat{m}}{\partial \tau^d} = -\frac{\partial \Delta V / \partial \tau^d}{\Delta V_m}, \quad \frac{\partial \widehat{m}}{\partial \tau^a} = -\frac{\partial \Delta V / \partial \tau^a}{\Delta V_m}, \quad \frac{\partial m_p}{\partial \tau^d} = -\frac{\partial V / \partial \tau^d}{V_m}, \quad \frac{\partial m_p}{\partial F_d} = \frac{1}{V_m}.$$

By envelope, $\partial V(m, 1 + \tau^r) / \partial \tau^r = (p^* - m) p^* D'((1 + \tau^r) p^*) < 0$, since $p^* > m$ (Lemma 1) and $D' < 0$ (D1). Hence $\partial \Delta V / \partial \tau^d = -\partial V(m, 1 + \tau^d) / \partial \tau^d > 0$ and $\partial \Delta V / \partial \tau^a = \partial V(m, 1 + \tau^a) / \partial \tau^a < 0$. Combined with $\Delta V_m < 0$ and $V_m < 0$, the four signs in the proposition follow. $\square$

A.2.6 Proof of Corollary 2

Proof. Specialize the decomposition (6) of Section 2 to $K = 2$ exit margins, $k \in \{a, p\}$, with $\theta^a = \Pr(m < \widehat{m})$, $\theta^p = \Pr(m > m_p)$, $\bar{\beta}_a \equiv \mathbb{E}[\beta(m) \mid m < \widehat{m}]$, $\bar{\beta}_p \equiv \mathbb{E}[\beta(m) \mid m > m_p]$, and $\beta_{\text{survivor}} = \mathbb{E}[\beta(m) \mid \widehat{m} \leq m \leq m_p]$ from Proposition 4. Then

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \theta^a (\beta_{\text{survivor}} - \bar{\beta}_a) + \theta^p (\beta_{\text{survivor}} - \bar{\beta}_p). \quad (\text{A.2})$$

Sign of each term. By Proposition 3, low-cost firms have lower passthrough under Kimball, so $\bar{\beta}_a < \beta_{\text{survivor}}$ and the avoidance contribution $\theta^a (\beta_{\text{survivor}} - \bar{\beta}_a)$ is positive. High-cost firms have higher passthrough, so $\bar{\beta}_p > \beta_{\text{survivor}}$ and the participation contribution $\theta^p (\beta_{\text{survivor}} - \bar{\beta}_p)$ is negative.

Dominance. The survivor-cohort gap is positive, $\beta_{\text{survivor}} > \beta_{\text{cohort}}$, if and only if the magnitude of the positive contribution exceeds that of the negative contribution:

$$\theta^a (\beta_{\text{survivor}} - \bar{\beta}_a) > \theta^p (\bar{\beta}_p - \beta_{\text{survivor}}),$$

which is equation (11) in the body. Two special cases follow. If $\theta^p = 0$ (no participation exit), the avoidance margin dominates trivially. If $\theta^a = \theta^p$ and the within-region passthrough gaps are equal, the two margins offset exactly and $\beta_{\text{survivor}} = \beta_{\text{cohort}}$. $\square$

A.3 MVPF Derivation

This appendix derives equation (8) for a marginal change in the tax rate τ evaluated at the counterfactual allocation.

The planner chooses τ to maximize welfare subject to a revenue constraint:

$$\max_{\tau} W(\tau) \quad \text{subject to} \quad \bar{R} = R(\tau). \quad (\text{SPP})$$

The MVPF is the shadow value of relaxing the constraint, $\text{MVPF} = (-\partial W / \partial \tau) / (\partial R / \partial \tau)$.

By the consumer envelope, the buyer's first-order welfare loss per unit pre-tax base is $\beta_i d\tau$: the price increase times quantity, with the buyer's own optimization vanishing. This holds across market structures (Weyl and Fabinger 2013). The seller's first-order loss is $(1-\beta_i) d\tau$ under perfect competition; under imperfect competition it carries an additional markup-interaction term, so the two sides need not sum to one (Weyl and Fabinger 2013). For exiters, I price the burden at the loss they would have borne had they remained in the direct channel. Aggregating under uniform within-group planner weights λ_B and λ_S , the resulting numerator is

$$N_{\text{exit}} = B^{\text{cohort}} [\lambda_B \beta_{\text{cohort}} + \lambda_S (1 - \beta_{\text{cohort}})],$$

where $\beta_{\text{cohort}} = (1 - \theta) \beta_{\text{survivor}} + \theta \bar{\beta}_{\text{exiter}}$. The welfare results in Section 6.2 use the nationalist specification, $\lambda_S = 0$, so only the buyer burden enters and the seller benchmark is not used in any reported figure. For the counterfactual exercise, I hold the pre-tax value of the cohort base fixed and let the observed tax base vary through exit from the direct channel. Exit need not remove a unit from revenue entirely: an exited unit may remain taxed at a lower rate $\tau' \leq \tau$, so revenue is $R(\tau) = [(1 - \theta(\tau)) \tau + \theta(\tau) \tau'] B^{\text{cohort}}$. Differentiating in τ with τ' fixed,

$$\frac{\partial R}{\partial \tau} = (1 - \theta) B^{\text{cohort}} - (\tau - \tau') \frac{\partial \theta}{\partial \tau} B^{\text{cohort}}.$$

With $\psi \equiv \partial \ln \theta / \partial \ln \tau$ the elasticity of the exit share with respect to the tax rate, $\tau \partial \theta / \partial \tau = \theta \psi$, so $(\tau - \tau') \partial \theta / \partial \tau = \theta \psi (1 - \tau' / \tau)$ and

$$\frac{\partial R}{\partial \tau} = B^{\text{cohort}} (1 - \theta) \left[1 - \frac{\theta}{1 - \theta} \psi \left(1 - \frac{\tau'}{\tau} \right) \right] = B^{\text{cohort}} (1 - \theta) (1 + \eta),$$

with $\eta = -(\theta / (1 - \theta)) \psi (1 - \tau' / \tau)$. The externality is negative whenever exit rises with the tax, and the denominator vanishes at $\eta = -1$. Setting $\tau' = 0$ recovers $\eta = \partial \ln(1 - \theta) / \partial \ln \tau = -(\theta / (1 - \theta)) \psi$, the elasticity of the surviving base.

Taking the ratio $N_{\text{exit}} / (\partial R / \partial \tau)$,

$$\text{MVPF}_{\text{exit}} = \frac{\lambda_B \beta_{\text{cohort}} + \lambda_S (1 - \beta_{\text{cohort}})}{(1 - \theta) (1 + \eta)},$$

which is equation (8); B^{cohort} cancels.

The numerator and denominator use different bases. The numerator prices burden on the full cohort B^{cohort} , including exiters at the loss they would have borne had they stayed in the direct channel. The denominator collects τ from the surviving base and τ' from the exited base. Exit therefore enters twice: the level θ shifts units from the rate τ to the lower rate τ' , and the response

$\partial\theta/\partial\tau$ changes marginal revenue.

At equal weights, $\lambda_B = \lambda_S = 1$, the competitive numerator is one and the MVPF is $1/[(1 - \theta)(1 + \eta)]$. It exceeds one whenever exit removes part of the base. At $\eta = 0$, the mechanical factor $1/(1 - \theta)$ already raises the MVPF above one; a negative fiscal externality raises it further. The equal-weights MVPF at the survivor cohort ($\beta_{\text{cohort}} = 1$), $1/[(1 - \theta)(1 + \eta)]$, is the lower endpoint of the negative-covariance range in Table A.2: 1.41 in aggregate, 1.51 for capital, 1.13 for consumption, and 2.10 for intermediates. The implied deadweight component is that value minus one.

The regression of Section 5.1.2 identifies the wedge-derivative $\partial\theta/\partial\Delta$, with $\Delta = \log[(1 + \tau)/(1 + \tau^h)]$. In general,

$$\psi = \frac{\partial \ln \theta}{\partial \ln \tau} = \frac{1}{\theta} \frac{\partial \theta}{\partial \Delta} \frac{\partial \Delta}{\partial \ln \tau}.$$

The hub rate τ^h does not move with the direct tariff τ , so $\partial\Delta/\partial \ln \tau = \tau/(1 + \tau)$, which is the conversion used in Section 6.2.

When exit decomposes into K mutually exclusive margins as in equation (6), each leaving revenue at its own rate τ'_k ,

$$\psi = \sum_{k=1}^K \frac{\theta_k}{\theta} \psi_k, \quad \psi_k \equiv \frac{\partial \ln \theta_k}{\partial \ln \tau},$$

and the fiscal externality decomposes as

$$\eta = - \sum_{k=1}^K \frac{\theta_k}{1 - \theta} \psi_k \left(1 - \frac{\tau'_k}{\tau}\right).$$

The application has two margins, and the reported MVPFs price each at the rate it actually pays. Transshipment removes a unit from the direct τ base but still clears under the hub's origin and pays the hub rate, so $\tau'_a = \tau^h$; shutdown removes the unit from trade and returns nothing, so $\tau'_p = 0$, with relocation folded into the same margin. When one margin dominates the aggregate is approximated by its contribution alone; counterfactual evaluation under composition-shifting policy requires the ψ_k separately.

A.3.1 Robustness of the Welfare Bound

Table A.2: Robustness of the Welfare Bound

End-use	θ^a	θ^p	MVPF, Cov < 0	MVPF, $\psi_p=\psi_a$	ψ multiple to Laffer
Capital	0.183	0.005	1.51–1.78	1.52	5.5×
Consumption	0.040	0.032	1.13–1.18	1.21	20.1×
Intermediate	0.218	0.009	2.10–2.56	2.16	2.6×
Aggregate	0.142	0.017	1.41–1.61	1.44	6.5×

Notes: High-screen 2023 values on the combined base $1 - \theta^a - \theta^p$ (both exit margins, as in Table A.3). The “MVPF, Cov < 0” column applies the negative-covariance cohort interval (using $\beta_{\max} = 2$ as the mirror of the $\beta_{\min} = 0$ floor) to equation (13). The “ $\psi_p=\psi_a$ ” column sets the shutdown elasticity equal to the transshipment elasticity, a generous upper bound on the participation fiscal externality, since firm exit is less tariff-elastic than rerouting. The final column reports the multiple of the estimated exit elasticity needed to reach $\eta_{\text{exit}} = -1$.

Table A.3: MVPF Robustness: Adding the Shutdown Margin (2023)

End-use	Exit shares			Fiscal externality		MVPF		
	$\bar{\theta}_{\text{low}}^a$	$\bar{\theta}_{\text{high}}^a$	$\bar{\theta}^p$	η_{exit} (low)	η_{exit} (high)	MVPF (low)	MVPF (mid)	MVPF (high)
Capital	0.061	0.183	0.005	-0.00	-0.18	1.01	1.14	1.51
Consumption	0.014	0.040	0.032	-0.01	-0.05	1.04	1.07	1.13
Intermediate	0.070	0.218	0.009	-0.05	-0.38	1.07	1.36	2.10
Aggregate	0.047	0.142	0.017	-0.01	-0.15	1.03	1.14	1.41

Notes: As Table 4, but adding the shutdown (participation) margin. $\bar{\theta}^p$ is the shutdown share, 2017-share-weighted; the two margins are disjoint by construction and enter the surviving base $1 - \theta^a - \theta^p$. The shutdown elasticity is set to zero (level effect only); Table A.2 reports the sensitivity to $\psi_p = \psi_a$.

Table A.4: Decomposing the Aggregate Exit-Based MVPF

Scenario	Exit-to-revenue conversion				MVPF accounting				
	θ^a	$\theta^a/(1 - \theta^a)$	ψ	η_{exit}	Nationalist numerator	Base factor	Fiscal factor	Nationalist MVPF	Equal-weight MVPF
Low	0.047	0.049	0.27	-0.01	0.95	1.05	1.01	1.01	1.06
Midpoint	0.094	0.104	0.67	-0.06	0.95	1.10	1.06	1.12	1.18
High	0.142	0.166	1.05	-0.15	1.00	1.17	1.18	1.37	1.37

Notes: Aggregate 2023 values. The low and high rows use the low and high transshipment screens from Table A.3; the midpoint row uses the mean transshipment share, the OLS midpoint slope, and $\beta_{\text{exiter}} = 0.5$. The exit elasticity is $\psi = (\partial\theta^a/\partial\Delta)(\partial\Delta/\partial\ln\tau)/\theta^a$. The fiscal externality nets the hub tariff still collected on transshipped units, $\eta_{\text{exit}} = -(\theta^a/(1 - \theta^a))\psi(1 - \tau^h/\tau^d)$. The nationalist numerator is β_{cohort} , the base factor is $1/(1 - \theta^a)$, and the fiscal factor is $1/(1 + \eta_{\text{exit}})$. The nationalist MVPF sets $\lambda_B = 1$ and $\lambda_S = 0$; the equal-weights MVPF sets $\lambda_B = \lambda_S = 1$.

B Screening Implementation Details

This appendix documents how I construct the exit share θ^a from trade flows consistent with tariff-induced transshipment through third countries.

B.1 Data and Notation

I use annual bilateral trade flows from CEPII’s BACI HS-12 vintage (2012–2023) at the HS6 product level. For each year t , country pair (i, j) , and HS6 code k , I observe the customs value of shipments $v_{i \rightarrow j, k, t}$.

For each HS6 product k and hub i in year t , define the two potential legs of transshipment:

- First leg (China $\rightarrow$ hub): $\ell_{k, i, t} \equiv v_{\text{CHN} \rightarrow i, k, t}$
- Second leg (hub $\rightarrow$ United States): $x_{k, i, t} \equiv v_{i \rightarrow \text{USA}, k, t}$

Define market shares as follows. Let $s_{k, i, t}^{\text{US}}$ denote country i ’s share of U.S. imports in product k :

$$s_{k, i, t}^{\text{US}} = \frac{v_{i \rightarrow \text{USA}, k, t}}{\sum_j v_{j \rightarrow \text{USA}, k, t}}.$$

Let $s_{k, i, t}^{\text{ROW}}$ denote country i ’s share of exports to the rest of the world (excluding the U.S.):

$$s_{k, i, t}^{\text{ROW}} = \frac{\sum_{j \neq \text{USA}} v_{i \rightarrow j, k, t}}{\sum_{j \neq \text{USA}} \sum_m v_{m \rightarrow j, k, t}}.$$

B.2 Hub Countries

I restrict potential hubs to a fixed list of 36 countries with geographic proximity to China, port infrastructure, and established trade relationships with both China and the U.S. (Table B.1). The

list is set before applying the screens and is not updated using post-2018 transshipment outcomes. This excludes implausible routes and reduces false positives from noise in global trade data.

Table B.1: Potential Transshipment Hubs (ISO-3 codes)

ARE	BHR	CAN	CHL	DEU	DOM	EGY	HKG	IDN	IND	ISR	JOR
JPN	KAZ	KHM	LAO	LKA	MAC	MAR	MEX	MYS	NLD	NPL	OMN
PAK	PAN	PHL	POL	RUS	SAU	SGP	THA	TUR	TWN	VNM	ZAF

Note: In BACI, Taiwan is recorded as “Other Asia” with country code S19.

B.3 Trend Estimation

I estimate product- and partner-specific trends in the pre-tariff period (2012–2017) with time centered at 2014.5.

Share trends (logit). For China and hub shares in U.S. imports and in ROW exports:

$$\text{logit}(s_{k,i,t}) = \alpha_{k,i} + \beta_{k,i}(t - 2014.5) + \varepsilon_{k,i,t}, \quad \widehat{s}_{k,i,t} = \text{invlogit}(\widehat{\alpha}_{k,i} + \widehat{\beta}_{k,i}(t - 2014.5)).$$

If the logit trend cannot be estimated because there are fewer than three pre-period observations or fewer than two shares strictly between zero and one, I use the pre-period mean $\bar{s}_{k,i,\text{pre}}$.

Level trends (log-linear). For trade levels $v \in \{\ell_{k,i,t}, x_{k,i,t}\}$:

$$\log(\max(v, 1)) = a_{k,i} + g_{k,i}(t - 2014.5) + v_{k,i,t}, \quad \widehat{v}_{k,i,t} = \exp(\widehat{a}_{k,i} + \widehat{g}_{k,i}(t - 2014.5)).$$

If estimation is unreliable, I use the pre-period mean $\bar{v}_{k,i,\text{pre}}$.

Surprise growth. Define above-trend growth, floored at zero:

$$\tilde{\ell}_{k,i,t} = \max\{\ell_{k,i,t} - \widehat{\ell}_{k,i,t}, 0\} \quad \text{and} \quad \tilde{x}_{k,i,t} = \max\{x_{k,i,t} - \widehat{x}_{k,i,t}, 0\}.$$

B.4 Screening Criteria

To be flagged as transshipment, a product-hub-year triplet (k, i, t) must simultaneously satisfy five conditions.

Screen 1: Tariff Exposure. Product k must have been exposed to U.S. tariff increases on China:

$$\tau_{k,2019}^d > \tau_{k,2017}^d,$$

where $\tau_{k,t}^d$ is the U.S. ad valorem tariff on Chinese product k in year t .

Screen 2: U.S. Reallocation. China's share of U.S. imports must fall below its pre-trend while the hub's share rises above its pre-trend:

$$s_{k,\text{CHN},t}^{\text{US}} < \widehat{s}_{k,\text{CHN},t}^{\text{US}} \quad \text{and} \quad s_{k,i,t}^{\text{US}} > \widehat{s}_{k,i,t}^{\text{US}}.$$

Screen 3: Rest-of-World Specificity. China's ROW export share must not fall relative to the hub's. Specifically, China's positive above-trend gain must weakly exceed the hub's positive above-trend gain:

$$\max\left\{s_{k,\text{CHN},t}^{\text{ROW}} - \widehat{s}_{k,\text{CHN},t}^{\text{ROW}}, 0\right\} \geq \gamma \max\left\{s_{k,i,t}^{\text{ROW}} - \widehat{s}_{k,i,t}^{\text{ROW}}, 0\right\},$$

where $\gamma = 1$ and the hub's positive gain must be nonzero. This rules out cases where the hub's global export performance in product k improves more than China's.

Screen 4: Tariff Feasibility. The U.S. tariff applied to product k when declared as originating in hub country i must be lower than the U.S. tariff on China:

$$\tau_{k,i,t}^{\text{hub}} < \tau_{k,t}^d.$$

Here $\tau_{k,i,t}^{\text{hub}}$ is the U.S. tariff on product k from hub country i , from Teti (2025). This ensures there is an economic incentive to transship through hub i .

Screen 5: Quantitative Consistency. The China-to-hub flow must be large enough to plausibly supply the hub-to-U.S. flow. I impose two versions:

(a) **Low screen (growth-based):** Uses above-trend growth on both legs. The condition is:

$$\tilde{\ell}_{k,i,t} \geq \rho^* \max(\tilde{x}_{k,i,t}, x_{\text{hinge},\Delta}).$$

The flagged quantity is the bottleneck, $\min(\tilde{\ell}_{k,i,t}, \tilde{x}_{k,i,t})$, so the low screen assigns positive exit only when both above-trend legs are positive.

(b) **High screen (levels-based):** Uses trade levels directly. The conditions are:

$$\min(\ell_{k,i,t}, x_{k,i,t}) \geq x_{\text{hinge,levels}} \quad \text{and} \quad \ell_{k,i,t} \geq \phi^* x_{k,i,t}.$$

The flagged quantity is the bottleneck: $\min(\ell_{k,i,t}, x_{k,i,t})$.

Parameter Calibration Table B.2 lists fixed parameters and calibrated thresholds.

Table B.2: Screening Parameters

Parameter	Value	Description
γ	1.0	ROW specificity multiplier (Screen 3).
$x_{\text{hinge},\Delta}$	0	Hinge for $\tilde{x}$ in growth rule (thousands of current USD).
$x_{\text{hinge,levels}}$	200	Minimum size for $\min(\ell, x)$ in levels rule (thousands of current USD).
ρ^*	0	Growth-rule proportionality threshold (Screen 5a).
ϕ^*	0.325	Levels-rule proportionality threshold (Screen 5b).

I calibrate ρ^* and ϕ^* by placebo discipline. For the placebo exercise, I hold out Screen 1 and apply Screens 2–5 to never-treated products, those with $\tau_{k,2019}^d = \tau_{k,2017}^d$. I choose the smallest grid values for which the implied exit share does not exceed 1% in any pre-tariff year from 2012 to 2017. This gives $\rho^* = 0$ and $\phi^* = 0.325$.

Aggregation and Capping For each product k and year t , I aggregate flagged values across hubs passing Screens 1–5:

$$Q_{k,t}^{\text{flag}} = \sum_{i \in \mathcal{H}} \mathbf{1}[\text{Screens 1–5 pass for } (k, i, t)] \times q_{k,i,t},$$

where $q_{k,i,t}$ is the flagged quantity from Screen 5, either $\min(\tilde{\ell}_{k,i,t}, \tilde{x}_{k,i,t})$ for the low-screen version or $\min(\ell_{k,i,t}, x_{k,i,t})$ for the high-screen version.

I deflate the aggregate to 2017 dollars and cap it at the 2017 China-to-U.S. baseline:

$$\theta_{k,t}^a = \min \left(\frac{Q_{k,t}^{\text{flag}} / P_t}{v_{\text{CHN} \rightarrow \text{USA}, k, 2017}}, 1 \right),$$

where P_t is the BLS end-use import price index for all commodities, normalized so $P_{2017} = 1$.

Persistence Filter I remove isolated one-year spikes by requiring flagged exit to appear in an adjacent year:

$$\theta_{k,t}^a \leftarrow \theta_{k,t}^a \times \mathbf{1} \left[Q_{k,t-1}^{\text{flag}} > 0 \text{ or } Q_{k,t+1}^{\text{flag}} > 0 \right].$$

B.5 Participation Exit Proxy

The participation margin is not a residual decline in direct trade. I construct a binary HS6-level proxy for products most likely to contain participation exit, then aggregate those flags with 2017 China-to-U.S. import weights. The proxy is conservative within flagged products: if an HS6 passes the screen, its full 2017 baseline value enters the numerator of θ^p , even though HS6 data do not reveal the exact firm-level share that shut down.

For each HS6 code k , let $v_{k,t}^{\text{US}}$ denote real China-to-U.S. trade and $v_{k,t}^{\text{ROW}}$ denote China's real exports to all non-U.S. destinations. I estimate log-linear 2012–2017 pre-trends for both series and define

$$r_{k,t} = \frac{v_{k,t}^{\text{US}}}{\max\{\widehat{v}_{k,t}^{\text{US}}, 1\}}, \quad r_{k,t}^{\text{ROW}} = \frac{v_{k,t}^{\text{ROW}}}{\max\{\widehat{v}_{k,t}^{\text{ROW}}, 1\}}.$$

Missing product-years are treated as zero trade before forming these ratios. Let A_k^s be an indicator that the transshipment screen $s \in \{\text{low}, \text{high}\}$ assigns positive θ^a to product k in any post-period year. The shutdown proxy for screen s is

$$\theta_k^{p,s} = \mathbf{1}\{r_{k,2018} < \epsilon_{\text{shut}}^*, r_{k,2019} < \epsilon_{\text{shut}}^*, r_{k,2023} < \epsilon_{\text{shut}}^*\} \mathbf{1}\{r_{k,2018}^{\text{ROW}} < 1, r_{k,2019}^{\text{ROW}} < 1, r_{k,2023}^{\text{ROW}} < 1\} \mathbf{1}\{A_k^s = 0\}.$$

The first indicator requires an immediate post-tariff collapse relative to trend and persistence through 2023. The second requires China's global exports of the product to fall below trend in the same years, which filters out U.S.-side substitution when Chinese exports to other destinations remain healthy. The third removes products with a detected transshipment footprint.

I calibrate ϵ_{shut}^* on never-treated placebo products using the U.S.-cessation gate alone. The chosen value is the largest grid point for which the value-weighted placebo shutdown share remains below one percent in every pre-tariff window; the resulting value is $\epsilon_{\text{shut}}^* = 0.825$. I then apply the rest-of-world gate at this fixed threshold. In the time-series figures and aggregate tables, I set $\theta_{k,t}^{p,s} = 0$ before 2018 and $\theta_{k,t}^{p,s} = \theta_k^{p,s}$ in every post-period year. The shutdown path is therefore a sustained post-period classification rather than a year-by-year re-estimation.

B.6 Tariff Data Sources

U.S. tariffs on China (τ^d) are from Amiti, Redding, and Weinstein (2020) for monthly HS10 rates through 2017, aggregated to annual HS6 using 2017 import shares as weights. From 2018–2023, I

use the Global Tariff Database from Teti (2025) and Rodríguez-Clare, Ulate, and Vasquez (2025), which provides bilateral HS6 tariffs. Hub country tariffs (τ^{hub}) are the U.S. tariff faced by each hub from the same source, namely the MFN rate for hubs without a U.S. trade agreement and the preferential rate for those with one.

C Empirical Results

This section reports supporting summary statistics and empirical checks for the exit measures and tariff-response estimates in the main text.

Table C.1: Summary Statistics

Variable	Pre		Post		Post-Pre	N
	Mean	SD	Mean	SD		
$\Delta_{k,t}$	0.008	0.059	0.089	0.079	0.082	4246
Consumption	0.008	0.032	0.050	0.064	0.042	942
Capital	0.010	0.128	0.124	0.070	0.114	608
Intermediate	0.006	0.026	0.146	0.067	0.140	2696
θ_{low}^a (transshipment, low screen)	0.001	0.013	0.028	0.075	0.027	4246
Consumption	0.001	0.008	0.010	0.032	0.009	942
Capital	0.002	0.016	0.043	0.074	0.041	608
Intermediate	0.001	0.019	0.054	0.118	0.052	2696
θ_{high}^a (transshipment, high screen)	0.009	0.066	0.100	0.204	0.091	4246
Consumption	0.006	0.044	0.028	0.078	0.022	942
Capital	0.016	0.087	0.159	0.230	0.144	608
Intermediate	0.011	0.084	0.207	0.287	0.196	2696
Share($\theta_{\text{low}}^a > 0$)	0.069	0.254	0.473	0.500	0.404	4246
Share($\theta_{\text{high}}^a > 0$)	0.071	0.257	0.469	0.500	0.399	4246
θ_{low}^p (shutdown, low screen)	0.000	0.000	0.013	0.113	0.013	4246
Consumption	0.000	0.000	0.017	0.131	0.017	942
Capital	0.000	0.000	0.007	0.081	0.007	608
Intermediate	0.000	0.000	0.008	0.090	0.008	2696
θ_{high}^p (shutdown, high screen)	0.000	0.000	0.014	0.116	0.014	4246
Consumption	0.000	0.000	0.018	0.134	0.018	942
Capital	0.000	0.000	0.004	0.067	0.004	608
Intermediate	0.000	0.000	0.010	0.101	0.010	2696
Consumption share (2017)	0.419	0.413	0.419	0.413	0.000	4246
Capital share (2017)	0.300	0.372	0.300	0.372	0.000	4246
Intermediate share (2017)	0.281	0.422	0.281	0.422	0.000	4246

Notes: Means and standard deviations are weighted by 2017 China-to-U.S. import exposure. Indented rows classify HS6 products by dominant 2017 end use. Regressions with end-use interactions use stacked fractional shares. Transshipment shares are the low and high screens of \$4.3. Cessation flags equal one only for sustained 2018, 2019, and 2023 declines with no detected transshipment footprint.

Table C.2: Transshipment Response to Tariffs

	Intensive Margin				Extensive Margin			
	θ_{low}^a		θ_{high}^a		$1(\theta_{\text{low}}^a > 0)$		$1(\theta_{\text{high}}^a > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.171*** (0.060)		0.573*** (0.186)		1.19*** (0.454)		1.22*** (0.427)	
$\Delta_{k,t} \times \text{Cap}$		0.134* (0.072)		0.492** (0.244)		0.901* (0.511)		0.956** (0.477)
$\Delta_{k,t} \times \text{Con}$		0.118*** (0.036)		0.254** (0.106)		0.940 (0.587)		0.903 (0.582)
$\Delta_{k,t} \times \text{Int}$		0.410*** (0.076)		1.43*** (0.153)		2.76*** (0.348)		2.79*** (0.355)
R^2	0.41	0.41	0.50	0.51	0.51	0.52	0.51	0.51
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year $\times$ Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports coefficients from equation (12). Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year $\times$ end-use fixed effects, weights by pre-period end-use share times the HS6 code’s China-to-U.S. import share, and HS6-clustered standard errors.

C.1 Pre-Trends

Figure 2 suggests that the tariffs raised transshipment, but aggregate time series cannot establish causality: transshipment might have been rising for unrelated reasons, or the screens might mechanically flag any trade reallocation.

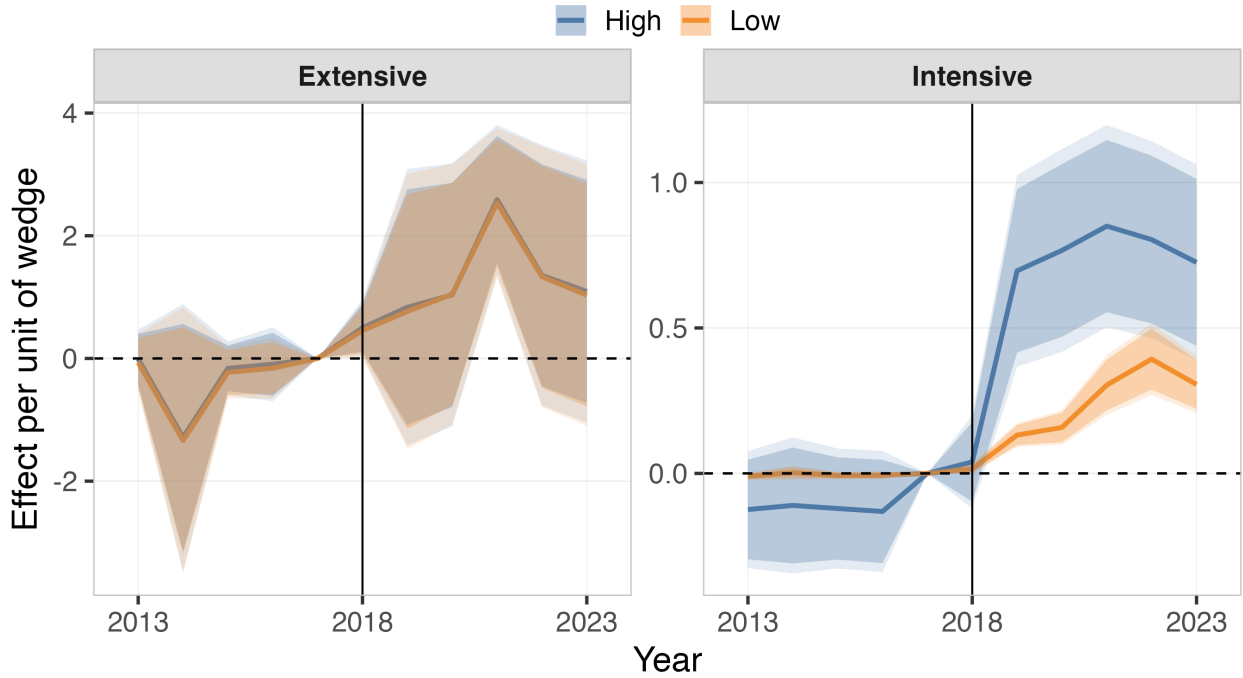
For each HS6 code k , I define exposure as the product’s 2019 tariff wedge, $\Delta_{k,2019}$, the first

full-implementation year.²¹ The event-study specification is

$$\theta_{k,t}^a = \alpha_k + \delta_{t \times e(k)} + \sum_{r \neq -1} \beta_r \mathbf{1}\{\text{rel}_t = r\} \times \Delta_{k,2019} + \varepsilon_{k,t}, \quad (\text{A.3})$$

where rel_t denotes years relative to 2018, α_k are HS6 fixed effects, and $\delta_{t \times e(k)}$ are year $\times$ end-use fixed effects. The outcome is $\theta_{k,t}^a$ on the intensive margin and $\mathbf{1}(\theta_{k,t}^a > 0)$ on the extensive margin. Observations are weighted by 2017 import shares and standard errors are clustered by HS6. Each β_r measures the difference in transshipment between high- and low-exposure products in year $2018 + r$, relative to 2017.

Figure C.1: Event Study: Transshipment Response to Tariff Exposure



Notes: Coefficients β_r from equation (A.3). Extensive margin (left) and intensive margin (right), with 90% and 95% confidence intervals. Low screen in orange; high screen in blue.

Figure C.1 plots the coefficients. Pre-trends are flat: from 2013 through 2017, intensive-margin coefficients are small and statistically indistinguishable from zero under both screens, and extensive-margin coefficients are noisy but centered near zero. Block-permutation tests confirm this for three of the four measures (Table C.3); the low-screen intensive margin shows a small pre-period gap ($p = 0.003$, Holm-adjusted). A direct pre-period interaction test yields small, in-

21. Anchoring exposure at 2019 pins coefficients to the response per unit of the long-run wedge rather than time-varying contemporaneous exposure. Results are robust to alternative anchoring years (Supplemental Appendix Figure S1).

significant coefficients (Supplemental Appendix Table S5).²² The post-2018 response is sharp on the intensive margin, with coefficients rising in 2019 and remaining elevated through 2023 under both screens. The extensive margin is less precisely estimated: post-period point estimates are positive, individually significant in 2021, and otherwise have confidence intervals that span zero.

Table C.3: Permutation Tests for Pre-Period Balance

Outcome	Observed Gap	P (left)	P (right)	P (two-sided)	P (Holm)
θ_{high}^a	0.007	0.967	0.033	0.065	0.196
θ_{low}^a	0.001	1.000	0.000	0.001	0.003
$\mathbf{1}\{\theta_{\text{high}}^a > 0\}$	-0.010	0.084	0.916	0.168	0.336
$\mathbf{1}\{\theta_{\text{low}}^a > 0\}$	0.001	0.131	0.869	0.261	0.336

Notes: Each row reports the pre-period treated-minus-placebo weighted mean gap and permutation-based p -values. The 5,000 block permutations reassign treated HS6 codes within HS4 groups while preserving the treated count. P (Holm) applies Holm–Bonferroni adjustment across outcomes. Outcomes are the low and high transshipment measures on the intensive and extensive margins.

C.2 Falsification and Selection Checks

The remaining appendix results probe the empirical design and the evidence used to assess the sign restriction in the main text. Table C.4 applies the transshipment screens to EU and Canada destinations as placebo outcomes. Figure C.2 permutes wedge trajectories across products, and Supplemental Appendix Figure S2 plots flagged transshipment for treated against never-treated codes in levels. Figure C.2 permutes wedge trajectories across products. Tables C.5–C.10 report heterogeneity, composition, and market-share checks tied to transshipment costs and composition artifacts.

22. Main-specification coefficients are also robust to residualizing HS6-specific pre-trends (Supplemental Appendix Table S6) and to collapsing the panel to long differences (Supplemental Appendix Table S7).

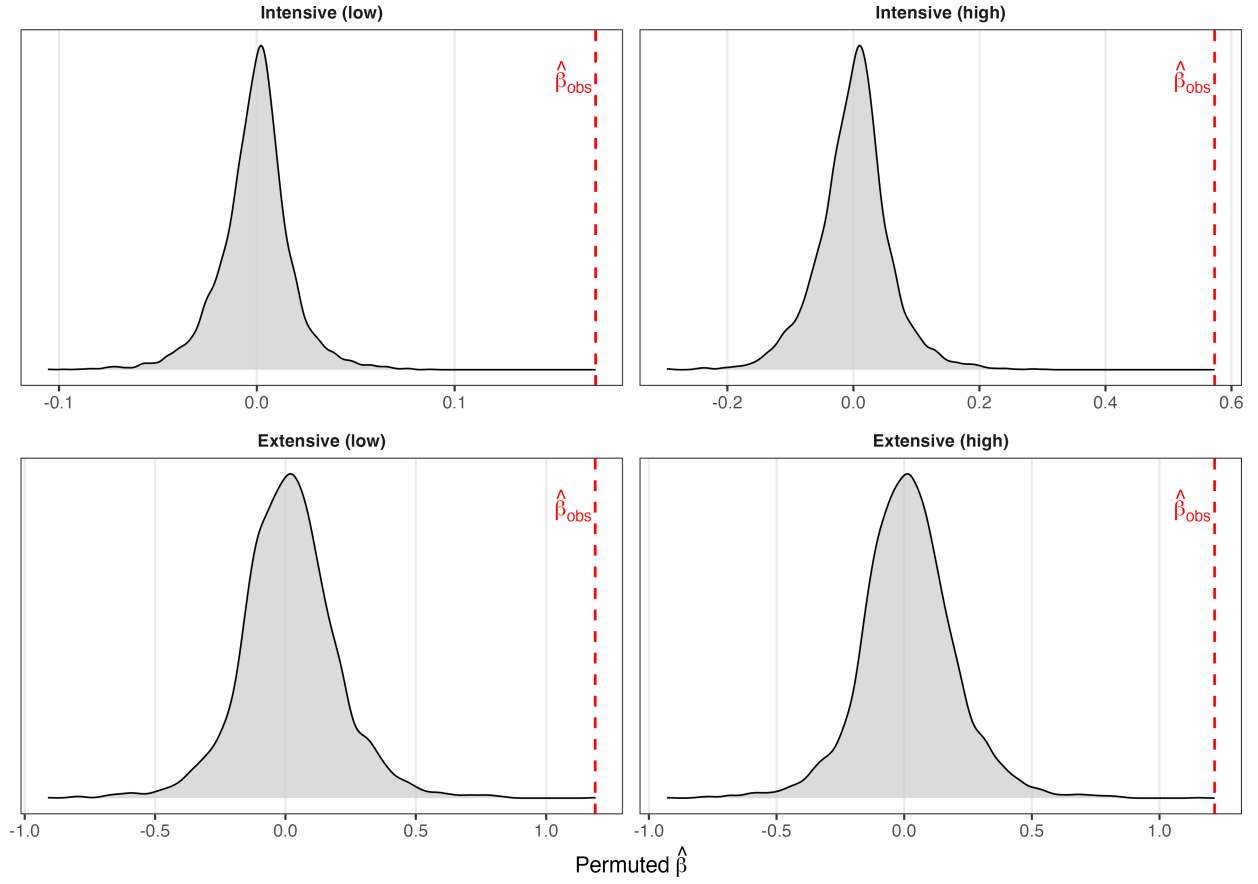
Table C.4: EU-27+UK and Canada Destination Placebo Results

EU-27+UK	θ_{low}^a		θ_{high}^a		$1(\theta_{\text{low}}^a > 0)$		$1(\theta_{\text{high}}^a > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.011 (0.009)	0.005 (0.004)	0.016 (0.013)	0.003 (0.005)	0.035 (0.076)	0.060 (0.066)	0.019 (0.079)	0.054 (0.065)
$\Delta_{k,t} \times 1(t \geq 2018)$		0.012 (0.012)		0.025 (0.019)		-0.049 (0.162)		-0.070 (0.169)
Observations	54,564	54,564	54,564	54,564	54,564	54,564	54,564	54,564
R ²	0.32	0.32	0.32	0.32	0.36	0.36	0.37	0.37
Canada	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.031** (0.014)	0.032 (0.022)	0.032* (0.017)	0.044 (0.033)	0.067 (0.083)	0.161 (0.110)	0.061 (0.071)	0.166 (0.125)
$\Delta_{k,t} \times 1(t \geq 2018)$		-0.004 (0.027)		-0.027 (0.042)		-0.202 (0.206)		-0.225 (0.200)
Observations	52,716	52,716	52,716	52,716	52,716	52,716	52,716	52,716
R ²	0.20	0.20	0.24	0.24	0.30	0.30	0.26	0.26
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Placebo transshipment measures are computed by replacing the U.S. destination with EU-27+UK (top panel) or Canada (bottom panel) and applying the same screening steps. The exposure gate uses the U.S. HS6 tariff panel, and the transshipment measure is capped within HS6 by the 2017 China→U.S. base. Columns (1)–(2) use θ_{low}^a , columns (3)–(4) use θ_{high}^a , columns (5)–(6) use $1\{\theta_{\text{low}}^a > 0\}$, and columns (7)–(8) use $1\{\theta_{\text{high}}^a > 0\}$. All specifications include HS6 and year×use fixed effects and cluster by HS6.

Figure C.2: Randomization Inference: Null Distributions of $\hat{\gamma}$ under Permuted Wedge Assignment



Notes: Each of 5,000 permutations randomly reassigns wedge trajectories across HS6 codes. Each product k inherits the full time series $\{\Delta_{k',t}\}_t$ from a donor product k' , preserving within-product wedge dynamics but randomizing which product faces which trajectory. Equation (12) is re-estimated with HS6 and year $\times$ end-use fixed effects, 2017 China $\rightarrow$ US import-share weights, and HS6 clustering. Gray densities plot the null distributions of $\hat{\gamma}$. Red dashed lines mark the observed $\hat{\gamma}$. Top panels use intensive-margin outcomes θ_{low}^a and θ_{high}^a . Bottom panels use extensive-margin outcomes $1(\theta_{\text{low}}^a > 0)$ and $1(\theta_{\text{high}}^a > 0)$.

Table C.5: Heterogeneous Fixed Costs Along the Intensive Margin

	θ_{low}^a		θ_{high}^a			
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,t}$	0.157*** (0.060)	0.162*** (0.057)	0.128*** (0.033)	0.481** (0.206)	0.506*** (0.182)	0.407*** (0.087)
$\Delta_{k,t} \times \text{Pre-2017 Hub Intensity}$	0.053 (0.043)			0.353*** (0.116)		
$\Delta_{k,t} \times \text{Bulkiness}$		-0.040 (0.028)			-0.287** (0.142)	
$\Delta_{k,t} \times \text{Electronics}$			0.100*** (0.025)			0.379*** (0.063)
R ²	0.41	0.41	0.42	0.51	0.50	0.53
Observations	54,912	54,912	54,912	54,912	54,912	54,912
Weighted	✓	✓	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each column regresses $\theta_{k,t}^a$ on the effective wedge and its interaction with one pre-2017 product characteristic. Hub intensity is the standardized pre-2017 hub share of China's exports of product k ; bulkiness is kilograms per U.S. dollar; electronics is an indicator for HS2 codes 84, 85, or 90. All specifications include HS6 and year×use fixed effects, weight by 2017 China→US exposure times pre-period end-use share, and cluster by HS6.

Table C.6: Heterogeneous Fixed Costs Along the Extensive Margin

	$1(\theta_{\text{low}}^a > 0)$			$1(\theta_{\text{high}}^a > 0)$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,t}$	1.14** (0.449)	1.20*** (0.438)	1.04*** (0.392)	1.15*** (0.430)	1.20*** (0.414)	1.07*** (0.366)
$\Delta_{k,t} \times \text{Pre-2017 Hub Intensity}$	0.181 (0.328)			0.245 (0.322)		
$\Delta_{k,t} \times \text{Bulkiness}$		0.068 (0.249)			-0.078 (0.255)	
$\Delta_{k,t} \times \text{Electronics}$			0.344* (0.188)			0.333* (0.185)
R ²	0.51	0.51	0.51	0.51	0.51	0.51
Observations	54,912	54,912	54,912	54,912	54,912	54,912
Weighted	✓	✓	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each column regresses the indicator for positive transshipment on the effective wedge and its interaction with one pre-2017 product characteristic. Hub intensity is the standardized pre-2017 hub share of China's exports of product k ; bulkiness is kilograms per U.S. dollar; electronics is an indicator for HS2 codes 84, 85, or 90. All specifications include HS6 and year×use fixed effects, weight by 2017 China→US exposure times pre-period end-use share, and cluster by HS6.

Table C.7: Decomposition of the Change in HS6-Average China Unit Value: Composition vs. Within-Variety

Year-pair	HS6	ΔP (%)	Comp. (%)	Within (%)	Comp. share	Within share
2016-2017	1828	0.058	-0.002	0.024	2.8	38.2
2017-2018	1885	0.040	0.000	0.024	0.6	60.7
2018-2019	1910	0.058	0.005	0.034	9.1	58.3
<i>By end-use, 2018–2019 pair:</i>						
Capital	386	0.151	0.063	0.040	42.1	26.9
Intermediate	1004	0.070	0.055	0.021	66.9	25.7
Consumption	520	-0.019	-0.059	0.034	59.6	34.1

Notes: Change in the value-weighted HS6-average China→US HS10 customs unit value, decomposed into composition, within-variety, and cross terms. Components are normalized by the prior-year HS6 price level and aggregated with 2017 China→US value weights. Entry and exit of HS10 varieties are assigned to composition. “Comp./Within share” is $|\cdot|/(|comp| + |within| + |cross|)$. The sample is Section 301 HS6 codes and Jan.–Oct. months.

Table C.8: Composition vs. Selection: Price Response to θ^a

	$\Delta \ln p_{CN}$			
	(1)	(2)	(3)	(4)
θ_{low}^a	-0.124 (0.114)		-0.095 (0.117)	
θ_{high}^a		-0.137** (0.053)		-0.130*** (0.050)
$\theta_{low}^a \times \text{UV dispersion}$			-0.056** (0.028)	
$\theta_{high}^a \times \text{UV dispersion}$				-0.034 (0.032)
R^2	0.09	0.09	0.09	0.09
Observations	46,474	46,474	46,474	46,474
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Regressions of the change in China unit values on θ^a (columns 1–2) and on θ^a interacted with standardized pre-tariff within-HS6 unit-value dispersion (columns 3–4). All specifications include HS6 and year×end-use fixed effects, weight by 2017 China→US exposure times pre-period end-use share, and cluster standard errors by HS6.

Table C.9: Survivor Passthrough by Exit Intensity: Quartile Binning versus Continuous Specifications

<i>Panel A: Quartile gradient</i>			
	$\Delta_{k,t} \times Q2$	$\Delta_{k,t} \times Q3$	$\Delta_{k,t} \times Q4$
Ties ordered by HS6 code (published)	2.255*** (0.694)	2.234*** (0.687)	1.612** (0.693)
Ties randomized (mean, 200 draws)	0.012	0.322	-0.423
<i>Panel B: Continuous and within-product specifications</i>			
	Continuous $\bar{\theta}^a$	Rank $\bar{\theta}^a$	Within-product $\theta_{k,t}^a$
$\Delta_{k,t} \times \theta^a$	-0.138 (0.777)	-0.438 (0.550)	-0.337 (1.120)
Observations		33,505	
HS6 & Year×Use FE	✓	✓	✓
Weighted	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Dependent variable is the gross-of-tariff log China unit value. The sample is differentiated (Rauch) HS6 codes facing U.S. tariff increases. Panel A bins products by post-2018 exit-intensity quartiles and reports interactions of the tariff wedge with Q2–Q4 under two tie-breaking rules. Panel B replaces quartiles with continuous post-2018 exit intensity, its rank, and time-varying $\theta_{k,t}^a$. All specifications include HS6 and year×end-use fixed effects, weight by 2017 China→US exposure times pre-period end-use share, and cluster standard errors by HS6.

Table C.10: Market-Share Co-Movement with Transshipment

	Low	High
	(1)	(2)
<i>Panel A: Extensive margin, $\mathbf{1}(\theta_{k,t}^a > 0)$</i>		
$\Delta_{k,t} \times \text{Pre-Tariff China Mkt. Share}$	0.874** (0.347)	0.898*** (0.346)
<i>Panel B: Intensive margin, $\theta_{k,t}^a$</i>		
$\Delta_{k,t} \times \text{Pre-Tariff China Mkt. Share}$	-0.087*** (0.033)	-0.070 (0.112)
<i>Panel C: Numerator, $\ln \max(\theta_{k,t}^a V_k^{2017}, 1)$</i>		
$\Delta_{k,t} \times \text{Pre-Tariff China Mkt. Share}$	11.644*** (4.171)	14.046*** (4.801)
Feasibility controls (hub, bulk, electronics)	✓	✓
HS6 & Year $\times$ Use FE	✓	✓
Weighted	✓	✓
Observations	26,316	26,316

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Each cell reports the coefficient on the tariff wedge interacted with standardized pre-tariff China share of U.S. imports. Panels A–C use the indicator for positive transshipment, transshipment share, and log flagged value. All regressions include wedge interactions with pre-2017 hub exposure, bulkiness, and electronics content; HS6 and year $\times$ end-use fixed effects; 2017 China $\rightarrow$ US exposure times pre-period end-use share weights; and HS6-clustered standard errors.

C.3 The Hub-Tariff Constraint on Transshipment

The wedge $\Delta_{k,t}$ moves with both the direct tariff and the hub tariff, and the estimates of Section 5.1.2 do not separate the two. Appendix S.1.4 derives the prediction for the hub component: holding the direct tariff fixed, products whose hubs face higher U.S. tariffs gain less from rerouting and should transship less. I therefore ask whether transshipment rises less after 2018 for products facing higher hub tariffs. The specification replaces the wedge in equation (12) with the

interaction of a post-2018 indicator and $\bar{\tau}_k^h$, the standardized 2018 value-weighted hub tariff, estimated on treated HS6 codes over 2012–2023. Observations are weighted by 2017 import shares and standard errors are clustered by HS6. The interaction measures the post-2018 difference in transshipment between high- and low-hub-tariff products, and a pre-period version interacts $\bar{\tau}_k^h$ with year indicators for 2012–2016.

Table C.11: Hub-Tariff Constraint on Transshipment: Difference-in-Differences with Pre-Trend Test

	Intensive margin		Extensive margin	
	θ_{low}^a	θ_{high}^a	$1(\theta_{\text{low}}^a > 0)$	$1(\theta_{\text{high}}^a > 0)$
	(1)	(2)	(3)	(4)
$\text{Post}_t \times \bar{\tau}_k^h$	−0.003	−0.009*	−0.052**	−0.049**
	(0.002)	(0.005)	(0.023)	(0.023)
Pre-trend test (p , 2012–16)	0.74	0.14	0.58	0.66
Observations	52,896			
HS6 & Year×Use FE	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Treated HS6 codes, 2012–2023. $\bar{\tau}_k^h$ is standardized, so coefficients are per standard deviation of the 2018 value-weighted hub tariff. $\text{Post}_t = 1$ for 2018–2023. The pre-trend row reports the joint p -value for the 2012–2016 event-study interactions. Regressions use 2017 China→US exposure weights times pre-period end-use shares and cluster by HS6.

Table C.11 reports results on both margins. On the intensive margin, the transshipped share falls by 0.3 and 0.9 percentage points for a one-standard-deviation higher hub tariff under the low and high screens. On the extensive margin, the same increase lowers the probability that a product transships at all by 5.2 and 4.9 percentage points. The pre-period interactions are jointly indistinguishable from zero (p between 0.14 and 0.74), so products facing higher hub tariffs were not transshipping differently before the tariffs. The transshipment response therefore runs through the hub tariff as well as the direct tariff.

Supplemental Appendix

S.1 Supplemental Theoretical Results

S.1.1 Cross-Side Equivalence

The baseline model assigns channel-choice and price decisions to the exporter. In practice, U.S. importers and trade intermediaries also have incentives to arbitrage across direct and transshipped channels. The following lemma shows that the channel partition characterized in Proposition 4 is invariant to which side of the transaction bears the fixed costs, provided the parties bargain efficiently.

Lemma 2 (Cross-side equivalence). *Consider an exporter with marginal cost m and a U.S. importer who buys the variety and resells competitively to U.S. consumers at the delivered price $(1 + \tau^r)p$. Suppose the parties engage in efficient bargaining over channel choice $r \in \{d, a\}$, the operating decision, and the price p , with any protocol that selects an outcome on the joint-surplus frontier. Define joint surplus by*

$$J(m, 1 + \tau^d) \equiv V(m, 1 + \tau^d) - F_d, \quad J(m, 1 + \tau^a) \equiv V(m, 1 + \tau^a) - F_d - F_a.$$

The cutoffs $\widehat{m}$ and m_p of Proposition 4 satisfy

$$\Delta V(\widehat{m}) = F_a, \quad V(m_p, 1 + \tau^d) = F_d,$$

and depend only on $J(\cdot)$. The partition is therefore invariant to how F_a and F_d are allocated between exporter and importer.

Proof. Because the importer resells competitively, importer profit per transaction is zero and joint variable surplus from operating in channel r at price p equals exporter variable profit $(p - m)D((1 + \tau^r)p \mid \mathbf{P})$. Fixed costs F_a and F_d enter joint surplus identically regardless of which party writes the check, since either party paying out of pocket reduces the surplus available to divide between them. Maximizing variable surplus over p in each channel yields $V(m, 1 + \tau^r)$, and subtracting the relevant fixed costs gives $J(m, 1 + \tau^r)$ as defined.

Under efficient bargaining, the parties select $(r, p, \text{operate})$ to maximize J . The decision to transship is $J(m, 1 + \tau^a) \geq J(m, 1 + \tau^d)$. Substituting,

$$V(m, 1 + \tau^a) - F_d - F_a \geq V(m, 1 + \tau^d) - F_d \iff \Delta V(m) \geq F_a,$$

yielding the avoidance cutoff $\Delta V(\widehat{m}) = F_a$. The decision to operate at all is $\max_r J(m, 1 + \tau^r) \geq 0$, which under Assumption 2 reduces to $V(m, 1 + \tau^d) \geq F_d$, yielding the participation cutoff $V(m_p, 1 + \tau^d) = F_d$. Both conditions reference only $J(\cdot)$ and are invariant to the cost split.

The bargaining protocol determines how the realized surplus J is divided between exporter and importer through transfers but does not affect the channel chosen or the cutoffs at which channel choice flips. $\square$

The lemma rests on efficient bargaining, which selects an outcome on the joint-surplus frontier. Under inefficient protocols, including double marginalization, hold-up, or take-it-or-leave-it offers that exclude positive-surplus trades, the partition may shift in level, but the existence of two opposing exit margins is preserved.

S.1.2 Heterogeneous Fixed Costs

This appendix allows both fixed costs, the transshipment setup cost F_a and the direct-channel operating cost F_d , to vary across firms or products. Both cutoffs in Proposition 4 then become firm-specific. The partition holds pointwise; the dominance condition of Corollary 2 continues to govern the sign of the survivor-cohort gap; the bound of Proposition 2 holds at the aggregate.

Corollary 4 (Heterogeneous fixed costs). *Let Assumptions D1–D5 hold, and let $(F_{a,i}, F_{d,i})$ be drawn from a continuous joint distribution G_F on $[F_{a,\min}, F_{a,\max}] \times [F_{d,\min}, F_{d,\max}]$, independent of m_i , with $F_{a,\min} \geq \Delta V(m_p(F_{d,\max}; \tau^d))$ so that Assumption 2 holds for every realization.²³*

- (i) **Firm-specific cutoffs and partition.** *For each realization (F_a, F_d) , Proposition 4 delivers cutoffs $\widehat{m}(F_a; \tau^d, \tau^a)$ solving $\Delta V(\widehat{m}) = F_a$ and $m_p(F_d; \tau^d)$ solving $V(m_p, 1 + \tau^d) = F_d$, with $\widehat{m} \leq m_p$. Firm i transships if $m_i < \widehat{m}_i \equiv \widehat{m}(F_{a,i}; \cdot)$, ships directly if $\widehat{m}_i \leq m_i \leq m_{p,i} \equiv m_p(F_{d,i}; \cdot)$, and exits otherwise. The comparative statics of Proposition 4 hold pointwise; additionally $\partial \widehat{m} / \partial F_a < 0$ and $\partial m_p / \partial F_d < 0$.*
- (ii) **Selection on each margin.** *Under Kimball curvature, direct-channel passthrough $\beta(m)$ is increasing in m . Survival in the direct sample $\sigma_i = \mathbf{1}\{\widehat{m}_i \leq m_i \leq m_{p,i}\}$ is a window indicator: it jumps from 0 to 1 at $\widehat{m}_i$ (the avoidance margin removes low- β firms) and from 1 to 0 at $m_{p,i}$ (the participation margin removes high- β firms). The two margins push the survivor-cohort gap in opposite directions, exactly as in Corollary 2.*
- (iii) **Aggregate covariance and bound.** *Let $\theta^a \equiv \Pr(m_i < \widehat{m}_i)$ and $\theta^p \equiv \Pr(m_i > m_{p,i})$, with $\theta = \theta^a + \theta^p$. When the transshipment margin dominates (Corollary 2), $\text{Cov}_\omega(\beta_i, \sigma_i) \geq 0$ and*

23. Since m_p is decreasing in F_d and ΔV is decreasing in m , $\Delta V(m_p(F_d; \tau^d))$ is increasing in F_d ; the binding case is $F_d = F_{d,\max}$.

the case-(i) bound of Proposition 2 applies:

$$\beta_{\text{cohort}} \in \left[(1 - \theta) \beta_{\text{survivor}} + \theta \beta_{\min}, \beta_{\text{survivor}} \right].$$

Under participation dominance, the case-(ii) bound applies instead. The dominance regime is determined by the joint distribution of (F_a, F_d) together with G , not by either fixed cost in isolation.

- (iv) **Cross-category heterogeneity.** If (F_a, F_d) is deterministic by end-use category k , all conclusions hold within k . Lower $F_{a,k}$ raises $\widehat{m}_k$ and increases θ_k^a ; lower $F_{d,k}$ raises $m_{p,k}$ and decreases θ_k^p . This extension lets fixed costs vary across categories without changing the sign logic. In the empirical application, I attribute the observed end-use ordering primarily to statutory tariff differences and effective wedges (Sections 5.1); category-specific fixed costs are an additional source of heterogeneity, not a required explanation.

Proof. For each realization (F_a, F_d) , Proposition 4 delivers existence, uniqueness, and the sign pattern of the cutoffs. The IFT gives $\partial \widehat{m} / \partial F_a = 1 / \Delta V_m < 0$ and $\partial m_p / \partial F_d = 1 / V_m < 0$ since both denominators are negative. For (ii), the indicator σ_i is a window in m_i for each realization of $(F_{a,i}, F_{d,i})$; the contributions of the two cut points to $\text{Cov}_\omega(\beta_i, \sigma_i \mid F_a, F_d)$ enter with opposite signs, with avoidance contributing positively and participation contributing negatively, by the same logic as Corollary 2. By the law of total covariance, $\text{Cov}_\omega(\beta_i, \sigma_i) = \mathbb{E}_{F_a, F_d}[\text{Cov}_\omega(\beta_i, \sigma_i \mid F_a, F_d)] + \text{Cov}_{F_a, F_d}(\mathbb{E}[\beta_i \mid F_a, F_d], \mathbb{E}[\sigma_i \mid F_a, F_d])$; under independence the second term vanishes since $\mathbb{E}[\beta_i \mid F_a, F_d] = \mathbb{E}[\beta_i]$ is constant in (F_a, F_d) , so the aggregate covariance is the expectation of conditional covariances and inherits their sign whenever they are uniformly signed. The bound from Proposition 2 then applies. Category-specific statements are immediate by conditioning on k . $\square$

Remarks. Independence of (F_a, F_d) from m is sufficient, not necessary. The results extend whenever the joint distribution preserves the dominance ordering: a positive association between F_a and m would strengthen dominance of the transshipment margin (high-cost firms face higher F_a and are even less likely to transship); symmetrically, a positive association between F_d and m would strengthen participation dominance (high-cost firms face higher F_d and are even more likely to shut down). A sufficiently strong reversal of either association could in principle flip the dominant margin.

S.1.3 Variable Transshipment Costs

Section 3.1 treats per-unit transshipment costs as part of the effective transshipment wedge. This appendix verifies that equivalence. Variable transshipment costs affect only the avoidance margin: the participation cutoff m_p is unchanged because per-unit transshipment costs do not enter direct-channel profit.

Suppose transshipment raises the exporter's marginal cost from m to $m + \Delta c$ in addition to facing the lower effective tariff τ^a . Optimized variable profit in the transshipment channel becomes $V(m + \Delta c, 1 + \tau^a)$. The avoidance cutoff $\widehat{m}_\Delta$ solves

$$V(\widehat{m}_\Delta + \Delta c, 1 + \tau^a) - F_a = V(\widehat{m}_\Delta, 1 + \tau^d).$$

The participation cutoff is m_p , defined as in Proposition 4 by $V(m_p, 1 + \tau^d) = F_d$. The corollary below covers the small- Δc regime in which the cost penalty of transshipment does not exceed the tariff benefit at any cost level; for very large Δc the transshipment channel can become uniformly dominated by direct shipping and the avoidance region collapses.

Corollary 5 (Selection under variable transshipment costs). *Under **D1–D4**, let $q^a(x)$ and $q^d(x)$ denote optimized quantities in the transshipped and direct channels at marginal cost x . Suppose Δc satisfies $q^a(m + \Delta c) > q^d(m)$ over $[\underline{m}, m_p]$, and that $V(\underline{m} + \Delta c, 1 + \tau^a) - F_a > V(\underline{m}, 1 + \tau^d)$ so the lowest-cost exporter strictly prefers transshipment. Then there exists a unique avoidance cutoff $\widehat{m}_\Delta(\tau^d, \tau^a, F_a, \Delta c) \in [\underline{m}, m_p]$ such that*

$$m < \widehat{m}_\Delta \Rightarrow \text{Transship}, \quad \widehat{m}_\Delta \leq m \leq m_p \Rightarrow \text{Direct}, \quad m > m_p \Rightarrow \text{Exit}.$$

The avoidance cutoff satisfies $\partial \widehat{m}_\Delta / \partial \tau^d > 0$, $\partial \widehat{m}_\Delta / \partial \tau^a < 0$, $\partial \widehat{m}_\Delta / \partial F_a < 0$, and $\partial \widehat{m}_\Delta / \partial \Delta c < 0$. The participation cutoff m_p is unaffected by Δc .

Proof. Define $\tilde{H}(m; \Delta c) \equiv V(m + \Delta c, 1 + \tau^a) - F_a - V(m, 1 + \tau^d)$. By the envelope theorem, $\partial \tilde{H} / \partial m = -q^a(m + \Delta c) + q^d(m)$. Under **D1–D4** and $\tau^a < \tau^d$, the transshipment channel has higher demand at any common marginal cost. For Δc small enough that $q^a(m + \Delta c) > q^d(m)$ over the relevant range, $\partial \tilde{H} / \partial m < 0$. The premise $\tilde{H}(\underline{m}; \Delta c) > 0$ combined with continuity implies a unique cutoff $\widehat{m}_\Delta$ at which $\tilde{H} = 0$. The comparative statics follow from the IFT applied at the cutoff. The participation cutoff m_p depends only on $V(\cdot, 1 + \tau^d)$ and F_d , which are unaffected by Δc . $\square$

The corollary establishes that the variable-cost extension preserves the three-region partition of Proposition 4, with $\widehat{m}_\Delta$ shifted leftward relative to the $\Delta c = 0$ benchmark while m_p remains fixed. The structural input that Section 2 requires, the sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$, is unchanged.

S.1.4 Heterogeneous Avoidance Wedges

Section 3.1 treats per-unit avoidance costs $\Delta c \geq 0$ as part of the effective transshipment wedge. This subsection develops the cross-product implications. Here τ_k^a denotes the effective tariff on the avoidance channel; empirically it is the hub's U.S. tariff τ_k^h (Section 4), so products differ in the wedge through both τ_k^d and τ_k^h . Within each product, only the avoidance cutoff $\widehat{m}_k$ varies with the wedge; the participation cutoff $m_{p,k}$ depends on τ_k^d and F_d , not on the avoidance channel.

Heterogeneous effective wedges. Let the effective avoidance wedge τ_k^a vary across products k . Within each product, τ_k^a is common across firms, so Proposition 4 applies directly: there exists a product-specific avoidance cutoff $\widehat{m}_k = \widehat{m}(\tau_k^d, \tau_k^a, F_a)$ solving $\Delta V_k(\widehat{m}_k) = F_a$ and a product-specific participation cutoff $m_{p,k} = m_p(\tau_k^d, F_d)$. Products with higher effective avoidance costs have smaller transshipment shares $\theta_k^a = G(\widehat{m}_k)$. Holding the participation margin fixed, the transshipment-margin lower bound used in the calibration is

$$\beta_{\text{cohort},k} \in \left[(1 - \theta_k^a) \beta_{\text{survivor},k} + \theta_k^a \beta_{\min}, \beta_{\text{survivor},k} \right].$$

The relevant policy variable is the effective wedge $\Delta_k \equiv \ln \frac{1+\tau_k^d}{1+\tau_k^h}$: products with a larger wedge have more avoidance.

S.2 Supplemental Empirical Results

Table S1: Transshipment Response to Tariffs (Clustered by HS4 $\times$ year)

	Intensive Margin				Extensive Margin			
	θ_{low}^a		θ_{high}^a		$1(\theta_{\text{low}}^a > 0)$		$1(\theta_{\text{high}}^a > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.171**		0.573**		1.19**		1.22**	
	(0.071)		(0.205)		(0.513)		(0.480)	
$\Delta_{k,t} \times \text{Cap}$		0.134		0.492*		0.901		0.956*
		(0.080)		(0.267)		(0.566)		(0.531)
$\Delta_{k,t} \times \text{Con}$		0.118**		0.254**		0.940		0.903
		(0.040)		(0.102)		(0.587)		(0.552)
$\Delta_{k,t} \times \text{Int}$		0.410***		1.43***		2.76***		2.79***
		(0.107)		(0.156)		(0.387)		(0.396)
R^2	0.41	0.41	0.50	0.51	0.51	0.52	0.51	0.51
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year $\times$ Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports coefficients from equation (12). Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year $\times$ end-use fixed effects, weights by pre-period end-use share times the HS6 code’s China-to-U.S. import share, and standard errors clustered by HS4 and year.

Table S2: Transshipment Response to Tariffs (unweighted)

	Intensive Margin				Extensive Margin			
	θ_{low}^a		θ_{high}^a		$1(\theta_{\text{low}}^a > 0)$		$1(\theta_{\text{high}}^a > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.028***		0.050***		0.143***		0.119***	
	(0.006)		(0.013)		(0.024)		(0.022)	
$\Delta_{k,t} \times \text{Cap}$		0.031***		0.034		0.113***		0.074*
		(0.012)		(0.039)		(0.041)		(0.044)
$\Delta_{k,t} \times \text{Con}$		0.011***		0.023***		0.100***		0.094***
		(0.004)		(0.008)		(0.026)		(0.025)
$\Delta_{k,t} \times \text{Int}$		0.040***		0.080***		0.191***		0.162***
		(0.012)		(0.022)		(0.050)		(0.042)
R^2	0.24	0.24	0.28	0.28	0.32	0.32	0.34	0.34
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted								

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports coefficients from equation (12). Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year $\times$ end-use fixed effects, no observation weights, and HS6-clustered standard errors.

Table S3: Transshipment Response to the Direct Tariff Alone

	θ_{low}^a		θ_{high}^a	
	(1)	(2)	(3)	(4)
$(1 + \tau_{k,t}^d)$	0.143*** (0.044)		0.501*** (0.141)	
$(1 + \tau_{k,t}^d) \times \text{Cap}$		0.128** (0.062)		0.463** (0.207)
$(1 + \tau_{k,t}^d) \times \text{Con}$		0.103*** (0.031)		0.236*** (0.090)
$(1 + \tau_{k,t}^d) \times \text{Int}$		0.281*** (0.053)		1.19*** (0.143)
R ²	0.40	0.41	0.49	0.50
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Specifications use the intensive-margin outcomes with the gross direct tariff $(1 + \tau_{k,t}^d)$ as the regressor in place of the two-route wedge $\Delta_{k,t}$. Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year × end-use fixed effects, weights by pre-period end-use share times the HS6 code’s China-to-U.S. import share, and HS6-clustered standard errors.

Table S4: Transshipment Response to Tariffs (with HS4×Year clustering and HS6, year, and HS2×Year Fixed Effects)

	Intensive Margin				Extensive Margin			
	θ_{low}^a		θ_{high}^a		$1(\theta_{\text{low}}^a > 0)$		$1(\theta_{\text{high}}^a > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.251**		0.897***		1.42**		1.44**	
	(0.084)		(0.236)		(0.544)		(0.526)	
$\Delta_{k,t} \times \text{Cap}$		0.190**		0.666**		1.33*		1.37*
		(0.081)		(0.256)		(0.694)		(0.675)
$\Delta_{k,t} \times \text{Con}$		0.173**		0.434**		1.07**		1.00**
		(0.065)		(0.185)		(0.468)		(0.450)
$\Delta_{k,t} \times \text{Int}$		0.356***		1.33***		1.63***		1.65***
		(0.095)		(0.194)		(0.436)		(0.429)
R ²	0.42	0.43	0.51	0.53	0.57	0.58	0.57	0.57
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year & Year×HS2 FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: This table reports coefficients from equation (12). Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6, year, and year × HS2 fixed effects, pre-period end-use-share weights, and standard errors clustered by HS4 and year.

Table S5: Pre-Period Test

	Intensive Margin		Extensive Margin	
	θ_{low}^a (1)	θ_{high}^a (2)	$1(\theta_{\text{low}}^a > 0)$ (3)	$1(\theta_{\text{high}}^a > 0)$ (4)
$\Delta_{k,t}$	0.265*** (0.037)	0.883*** (0.115)	1.73*** (0.448)	1.73*** (0.448)
$\Delta_{k,t} \times 1(t < 2018)$	-0.217*** (0.042)	-0.719*** (0.124)	-1.27** (0.505)	-1.19** (0.491)
R ²	0.41	0.51	0.52	0.51
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each column repeats the main intensive- or extensive-margin specification and adds the wedge interacted with a pre-2018 indicator. Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year × end-use fixed effects, weights by pre-period end-use share times the HS6 code’s China-to-U.S. import share, and HS6-clustered standard errors.

Table S6: Slope-adjusted difference-in-differences

	$\tilde{\theta}_{\text{low}}^a$	$\tilde{\theta}_{\text{high}}^a$	$1(\tilde{\theta}_{\text{low}}^a > 0)$	$1(\tilde{\theta}_{\text{high}}^a > 0)$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.165*** (0.058)	0.504*** (0.187)	1.19** (0.511)	1.22** (0.488)
R ²	0.40	0.47	0.51	0.50
Observations	54,912	54,912	54,912	54,912
Weighted	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each outcome is residualized by its HS6-specific linear trend estimated over 2012–2017 before estimation. The baseline specification is then re-estimated with HS6 and year×end-use fixed effects, 2017 China→US import-share weights multiplied by pre-period end-use share, and HS6-clustered standard errors.

Table S7: Long-differenced Specification

	θ_{Low}^a	θ_{High}^a	$1(\theta_{\text{low}}^a > 0)$	$1(\theta_{\text{high}}^a > 0)$	$1((\theta_{\text{low}}^a)^{\text{Act}} > 0)$	$1((\theta_{\text{high}}^a)^{\text{Act}} > 0)$
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,\text{Post}} - \Delta_{k,\text{Pre}}$	0.119*** (0.053)	0.396** (0.157)	0.761** (0.350)	0.813** (0.320)	0.682* (0.360)	0.723** (0.319)
R ²	0.15	0.24	0.10	0.11	0.14	0.14
Observations	4,246	4,246	4,246	4,246	2,626	3,064
Weighted	✓	✓	✓	✓	✓	✓
Use FE	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: For each HS6, pre means use 2012–2017 and post means use 2018–2023. Columns (1)–(4) regress long differences in the outcome on the long difference in the effective wedge with end-use fixed effects. Columns (5)–(6) restrict to HS6 codes with no pre-period activation. All regressions are weighted by 2017 import shares and cluster by HS6.

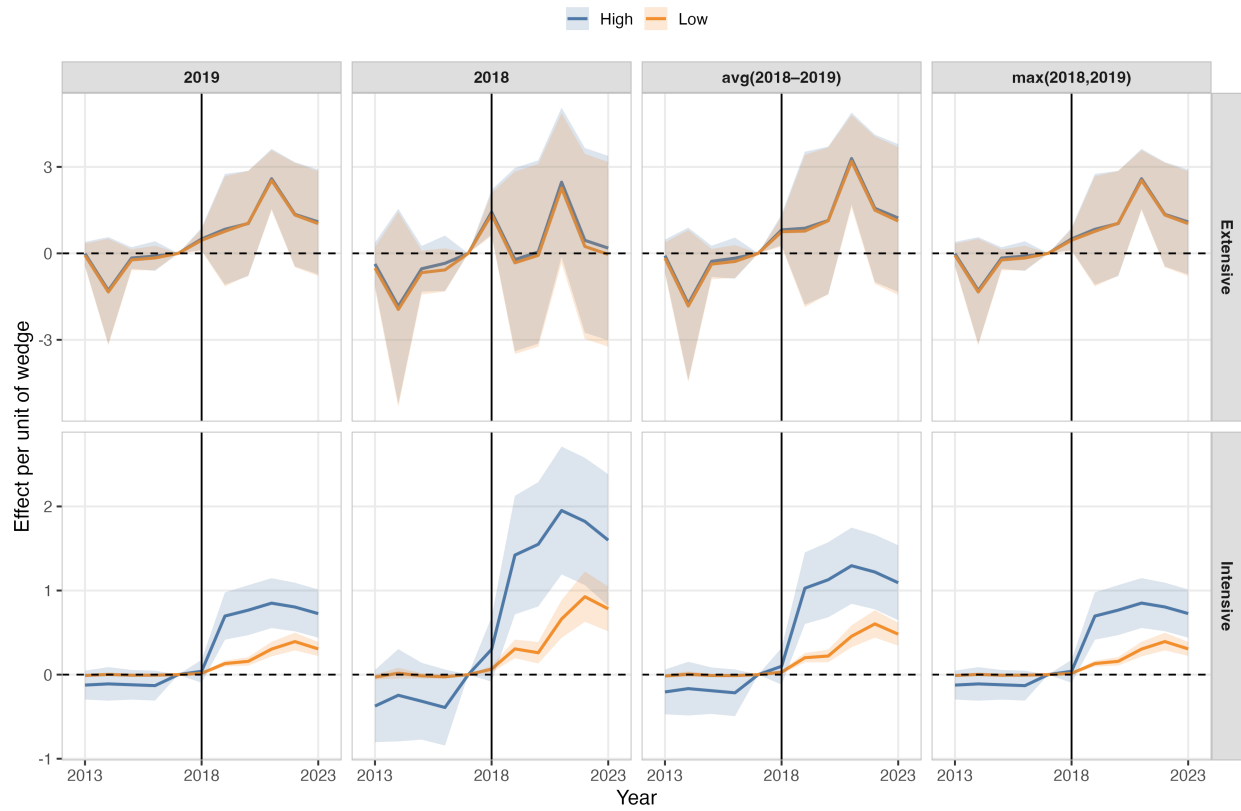
Table S8: Baseline Specification with HS4 Trends

	θ_{low}^a	θ_{high}^a	$1(\theta_{\text{low}}^a > 0)$	$1(\theta_{\text{high}}^a > 0)$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.133***	0.566***	1.75***	1.71***
	(0.032)	(0.141)	(0.608)	(0.616)
R ²	0.56	0.62	0.59	0.59
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓
HS4 Trend	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Specifications add linear trends for HS4 codes. Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year × end-use fixed effects, weights by pre-period end-use share times the HS6 code’s China-to-U.S. import share, and HS6-clustered standard errors.

Figure S1: Robustness to Anchoring Differently



Notes: Event-study estimates under four exposure anchors: the 2019 effective wedge, the 2018 wedge, the 2018–2019 average, and the maximum of the 2018 and 2019 wedges. Panels report extensive- and intensive-margin outcomes with the same fixed effects, weights, and clustering as equation (A.3).

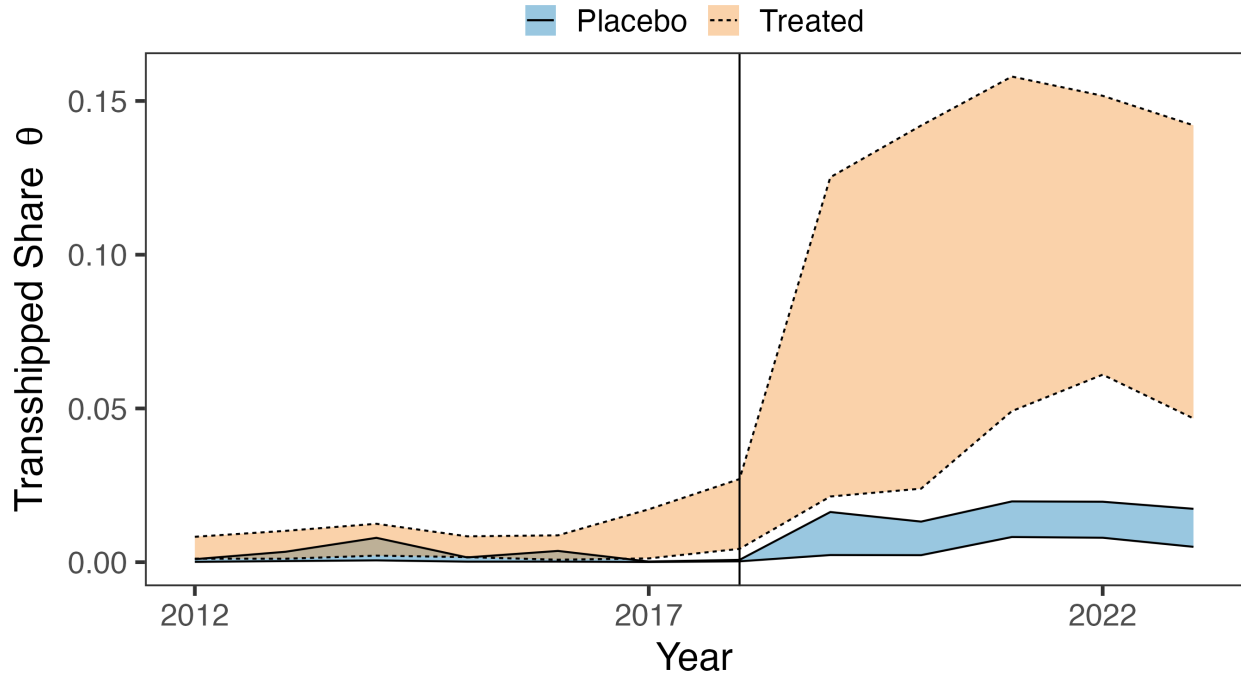
Table S9: Transshipment Response to Tariffs (2016-2019)

	Intensive Margin				Extensive Margin			
	θ_{low}^a		θ_{high}^a		$1(\theta_{\text{low}}^a > 0)$		$1(\theta_{\text{high}}^a > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.077***		0.473**		0.820		0.827	
	(0.030)		(0.186)		(0.582)		(0.585)	
$\Delta_{k,t} \times \text{Cap}$		0.056*		0.355*		0.517		0.515
		(0.032)		(0.211)		(0.545)		(0.545)
$\Delta_{k,t} \times \text{Con}$		0.057***		0.190**		0.477		0.491
		(0.018)		(0.093)		(0.966)		(0.972)
$\Delta_{k,t} \times \text{Int}$		0.206***		1.44***		2.72***		2.75***
		(0.033)		(0.197)		(0.372)		(0.380)
R^2	0.33	0.34	0.43	0.45	0.47	0.48	0.46	0.47
Observations	22,880	22,880	22,880	22,880	22,880	22,880	22,880	22,880
HS6 & Year $\times$ Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports coefficients from equation (12). Each HS6–year observation is fractionally allocated across end uses according to its UN-BEC shares. The stacked HS6–year–end-use panel includes HS6 and year $\times$ end-use fixed effects, weights by pre-period end-use share times the HS6 code’s China-to-U.S. import share, and HS6-clustered standard errors.

Figure S2: Aggregate Transshipment among Treated and Untreated HS6 Codes



Notes: Intensive-margin transshipment for treated and untreated HS6 codes, constructed as in Section B.

Table S10: The Entropy-Weighted Effect of Tariffs on Transshipment

	Intensive Margin		Extensive Margin	
	θ_{low}^a (1)	θ_{high}^a (2)	$1(\theta_{\text{low}}^a > 0)$ (3)	$1(\theta_{\text{high}}^a > 0)$ (4)
$\Delta_{k,t}$	0.170*** (0.060)	0.569*** (0.186)	1.18*** (0.453)	1.21*** (0.426)
R ²	0.41	0.50	0.51	0.51
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

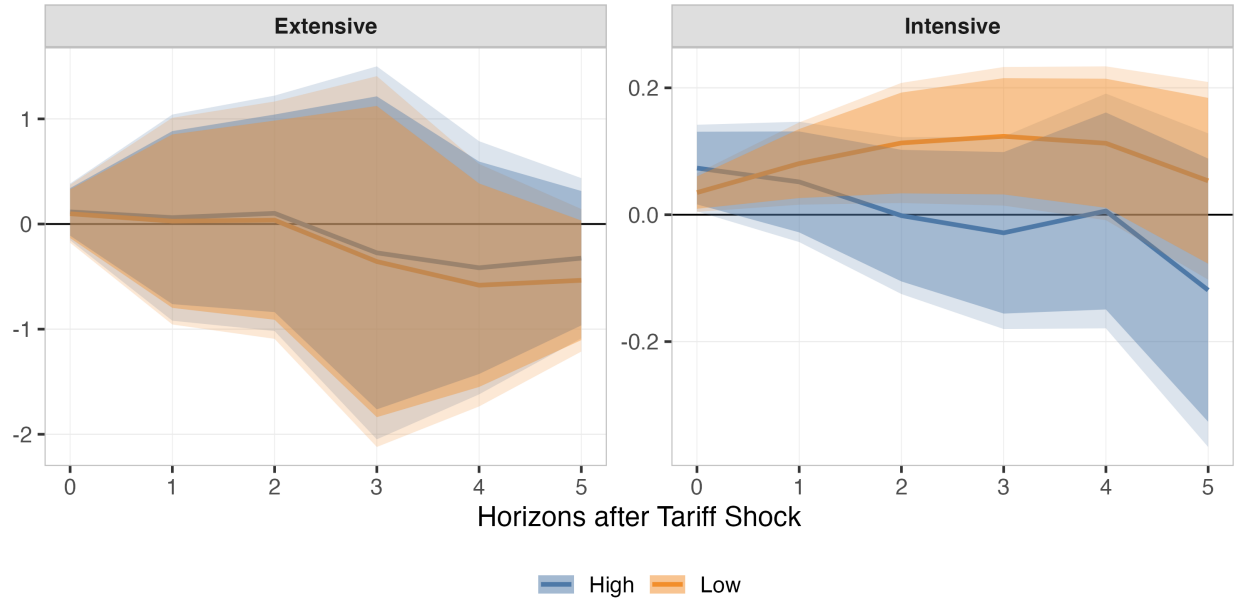
Notes: HS6 units are re-weighted by entropy balancing so treated and control units match pre-period transshipment moments. Regressions use 2017 exposure × entropy-balance weights × pre-period end-use share, HS6 and year×end-use fixed effects, and HS4-clustered standard errors. Coefficients are per unit of the wedge.

Table S11: Percent difference between baseline and leave-one-out: $100 \times \frac{\beta_{LOO} - \beta}{\beta}$

Country Code	Extensive (High)	Extensive (Low)	Intensive (High)	Intensive (Low)
CAN	-14.125	-15.589	16.957	18.119
MEX	5.650	6.440	31.304	25.370
TWN	-7.588	-8.560	13.291	20.744
DEU	-7.911	-6.315	21.977	13.164
SGP	-15.211	-14.480	2.427	3.749
JPN	-1.483	-2.459	28.953	27.882
HKG	-11.709	-12.462	4.973	4.358
VNM	-39.311	-38.999	0.333	4.638
IND	-2.502	-1.290	6.271	7.452
MYS	-2.788	-1.268	2.741	10.268
RUS	-14.019	-15.734	0.412	-0.345
ARE	-12.991	-13.964	-0.129	-0.208
POL	-1.195	-1.604	1.985	1.793
TUR	1.621	1.418	1.176	1.328
THA	-1.001	-0.456	3.034	6.753

Notes: For each regression specification, one hub is omitted and the coefficient is recomputed. Entries report the percent difference between the leave-one-out estimate and the baseline regression. High and Low denote the high and low transshipment screens. The table shows the fifteen hubs with the largest average absolute influence across outcomes. All underlying leave-one-out coefficients remain positive; the smallest absolute t -statistic is 2.87.

Figure S3: Local Projections of Transshipment from Tariff Exposure



Notes: Local-projection specification:

$$\theta_{k,t+h}^a - \theta_{k,t-1}^a = \alpha_k + \delta_{t \times e(k)} + \gamma_h \Delta_{k,t} + \varepsilon_{k,t+h}.$$

The specification is estimated separately for the extensive margin, using the linear probability model from the main text, and for the intensive margin in transshipment-share levels. Standard errors are clustered by HS6 code.

Table S12 relates measured exit intensity, in raw shares, to deviations of direct China–U.S. quantities and prices from their pre-tariff trends.

Table S12: Exit Intensity and Direct China→US Trade

	$\Delta^{\text{trend}}(\text{CN} \rightarrow \text{US})_{k,t}$			
	Quantities		Prices	
	(1)	(2)	(3)	(4)
θ_{low}^a	-1.06*** (0.283)		-0.877** (0.350)	
θ_{high}^a		-0.636*** (0.123)		-0.460*** (0.135)
R ²	0.46	0.47	0.34	0.35
Observations	53,485	53,485	53,134	53,134
HS6 FE	✓	✓	✓	✓
Year×Use FE	✓	✓		
Year FE			✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Regressions of $\Delta_{\text{CN} \rightarrow \text{US}}^{\text{trend}}$ —the deviation of HS6-level direct China→US imports from their 2012–2017 linear trend—on exit intensity. Columns 1–2: log import values, with HS6 and year×end-use fixed effects. Columns 3–4: China FOB unit values, with HS6 and year fixed effects. All specifications are weighted by pre-period China→US import shares; standard errors are clustered by HS6.